25 May 1982
SUPERSEDING
MIL-M-28787/45A(EC)
28 February 1977

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY CODED DECIMAL, DIGITAL, KEY CODE ZBZ

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary coded decimal counter module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN. ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.90 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZBZ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5 4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/45-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3 7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

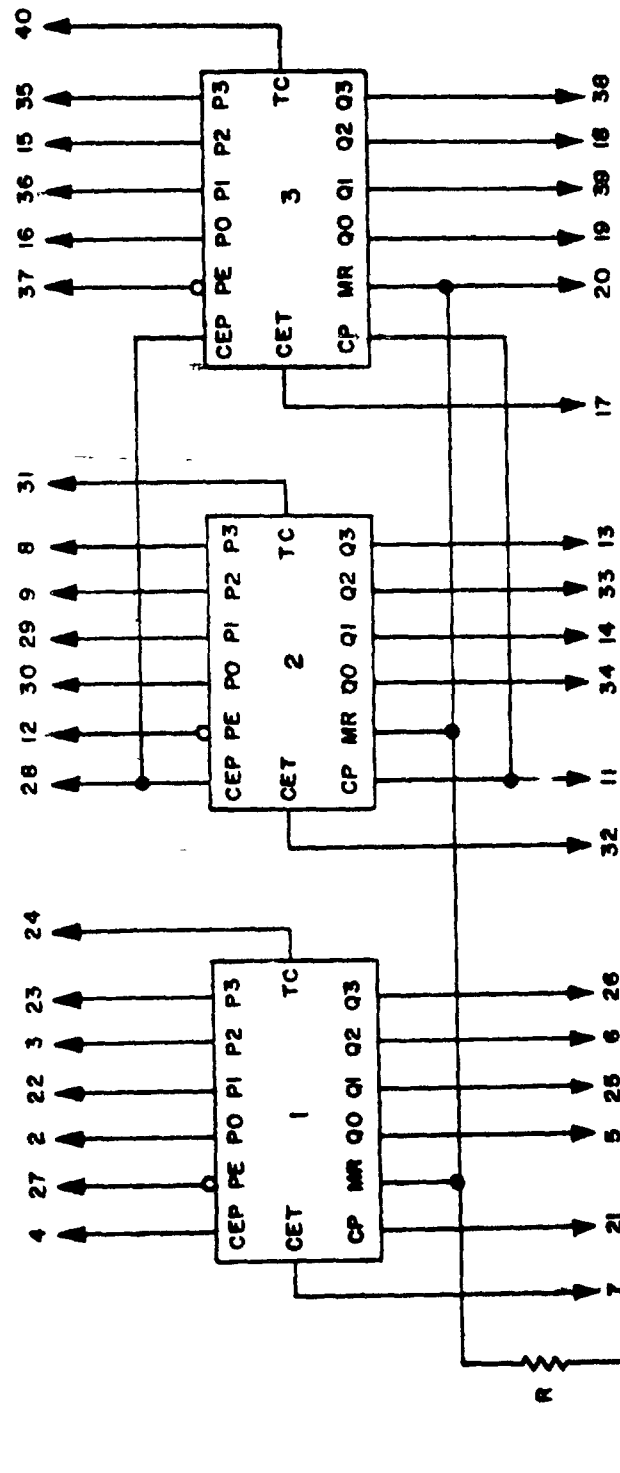
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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NOTES:
1. R = 10 kΩ ±5%, 1/4 W.
2. P0-P3 = Parallel inputs.
3. Q0-Q3 = Parallel outputs.
4. TC = Terminal count.
5. PE = Parallel enable input.
6. CP = Clock input.
7. CEP = Count enable parallel input.
8. CET = Count enable trickle input.
9. MR = Master reset input.

FIGURE 1. ZBZ module function diagram, binary coded decimal counter.

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TABLE I. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	V_{OUT} (low-loaded)	IV	S	0.01	0.60	0.01	0.60	0.01	0.60	0.01	0.60	V
B1-B20	Input breakdown current (PO-P3, CEP-1, CP-1, PE, and CET)	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
B21,B22	Input breakdown current (CEP-2, 3 and CP-2, 3)	V	S	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	mA
B23	nput breakdown current (MR-1, 2, 3)	V	S	0.0	3.0	0.0	3.0	0.0	3.0	0.0	3.0	mA
C1-C23	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D	Isolation	VII	S	10	-	1.0	-	10	-	1.0	-	MΩ
E1-E3	Functional count, high level	VIII	X	See table VIII								
F1-F6	Functional CET and PE, low level	VIII	X	See table VIII								
F7-F9	Functional CEP, low level	VIII	X	See table VIII								
G1-G15	Static output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H15	Static output voltage, low level	VIII	X	0.00	0.30	0.00	0.30	0.00	0.30	0.00	0.30	V
J1-J3	Input threshold check (MR), low level	VIII	X	See table VIII								
K1-K3	Input threshold check (CET), high level	VIII	X	See table VIII								
K4-K6	Input threshold check (CEP), high level	VIII	X	See table VIII								

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1	Input current, low level (MR)	X	X	0.82	1.88	0.82	1.88	0.82	1.88	0.82	1.88	mA
L2-L9	Input current, low level (CP-1, CET, PE, and CEP-2, 3)	X	X	0.10	0.86	0.10	0.86	0.10	0.86	0.10	0.86	mA
L10	Input current, low level (CP-2, 3)	X	X	0.10	1.72	0.10	1.72	0.10	1.72	0.10	1.72	mA
L11	Input current, low level (CEP-1)	X	X	0.10	0 43	0.10	0.43	0.10	0.43	0.10	0.43	mA
L12-L23	Input current, low level (PO-P3)	X	X	0.10	0 29	0.10	0.29	0.10	0.29	0.10	0.29	mA
M1	Input current, high level (MR)	X	X	-0 228	-0.336	-0.228	-0.336	-0.228	-0.336	-0.228	-0.336	mA
M2-M9	Input current, high level (CP-1, CET, PE, and CEP-2, 3)	X	X	0	40	0	40	0	40	0	40	µA
M10	Input current, high level (CP-2, 3)	X	X	0	80	0	80	0	80	0	80	µA
M11	Input current, high level (CEP-1)	X	X	0	20	0	20	0	20	0	20	µA
M12-M23	Input current, high level (PO-P3)	X	X	0.0	13 3	0.0	13.3	0.0	13 3	0.0	13.3	µA
N1-N15	Output short circuit current	XI	X	2.5	25	2.5	25	2.5	25	2.5	25	mA
P1,P2	Power supply current	XII	X	1.0	82 5	1.0	82.5	1.0	82.5	1.0	82.5	mA
Q1	Input capacitance (MR-1, 2, 3)	XIII	Q	3	45	3	45	3	45	3	45	pF
Q2,Q3	Input capacitance (CP-2, 3 and CEP-2, 3)	XIII	Q	2	30	2	30	2	30	2	30	pF
Q4-Q23	Input capacitance (all other inputs)	XIII	Q	2	20	2	20	2	20	2	20	pF

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R	Power supply transient amplitude	XIV	Q	0	400	0	400	0	400	0	400	mV
S1-S3	Propagation delay time (t_{PHL}), CP to TC, C_L = 40 pF	XV	S	10	49	10	49	10	52	10	52	ns
S4-S15	Propagation delay time (t_{PHL}), CP to Q, C_L = 40 pF	XV	S	15	59	15	59	15	62	15	62	ns
T1-T3	Propagation delay time (t_{PHL}), CP to TC, C_L = 100 pF	XV	Q	10	55	10	55	10	58	10	58	ns
T4-T15	Propagation delay time (t_{PHL}), CP to Q, C_L = 100 pF	XV	Q	15	65	15	65	15	68	15	68	ns
U1-U15	Transition time (t_{THL}), C_L = 40 pF	XVI	S	2	20	2	20	2	22	2	22	ns
V1-V15	Transition time (t_{THL}), C_L = 100 pF	XVI	Q	2	25	2	25	2	27	2	27	ns
W1-W3	Propagation delay time (t_{PLH}), CP to TC, C_L = 40 pF	XVII	S	20	99	20	99	20	105	20	105	ns
W4-W15	Propagation delay time (t_{PLH}), CP to Q, C_L = 40 pF	XVII	S	10	54	10	54	10	59	10	59	ns
X1-X3	Propagation delay time (t_{PLH}), CP to TC, C_L = 100 pF	XVII	Q	25	105	25	105	25	111	25	111	ns
X4-X15	Propagation delay time (t_{PLH}), CP to Q, C_L = 100 pF	XVII	Q	15	60	15	60	15	65	15	65	ns
Y1-Y15	Transition time (t_{TLH}), C_L = 40 pF	XVIII	S	5	45	5	45	5	50	5	50	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Z1-Z15	Transition time (t_{TLH}), C_L = 100 pF	XVIII	Q	10	55	10	55	10	60	10	60	ns
AA1-AA15	Master reset delay time (t_{PHL}), C_L = 40 pF	XIX	S	25	99	25	99	25	105	25	105	ns
BB1-BB15	Master reset delay time (t_{PHL}), C_L = 100 pF	XIX	Q	25	105	25	105	25	111	25	111	ns
CC1-CC3	Propagation delay time (t_{PHL}), CET to TC, C_L = 40 pF	XX	S	10	64	10	64	10	68	10	68	ns
CC4-CC6	Propagation delay time (t_{PLH}), CET to TC, C_L = 40 pF	XX	S	10	59	10	59	10	63	10	63	ns
DD1-DD3	Propagation delay time (t_{PHL}), CET to TC, C_L = 100 pF	XX	Q	15	70	15	70	15	75	15	75	ns
DD4-DD6	Propagation delay time (t_{PLH}), CET to TC, C_L = 100 pF	XX	Q	15	65	15	65	15	70	15	70	ns
EE1-EE3	Maximum counting frequency check (PW1)	XXI	S	225	275	225	275	225	275	225	275	ns
EE4-EE6	Maximum counting frequency check (PW2)	XXI	S	975	1,025	975	1,025	975	1,025	975	1,025	ns
FF1-FF6	Setup and hold time check (CET and CEP)	XXII	S	1.9	2.1	1.9	2.1	1.9	2.1	1.9	2.1	µs
GG1-GG12	Setup and hold time check (data), high level	XXIII	S	10	54	10	54	10	59	10	59	ns
GG13-GG27	Setup and hold time check (data and PE), low level	XXIII	S	10	59	10	59	10	62	10	62	ns

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
HH1-HH15	Output crosstalk, high level	XXIV	Q	0	400	0	400	0	400	0	400	mV
JJ1-JJ15	Output crosstalk, low level	XXIV	Q	0	500	0	500	0	500	0	500	mV
KK	Power supply filter capacitance	XXV	S	1.6	3 0	1.6	3 0	1.6	3.0	1.6	3.0	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through KK shall be performed in any order with the following exceptions. If test conditions A, B, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through KK in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C, E through P, and R through KK shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition KK, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C, E through P, and R through KK shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C and E through KK shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/45 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833637. Part number M28787/45-1 supersedes part numbers 2833637 and M28787/45. These parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I:

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Maximum power dissipation	455 mW (outputs not loaded)	
Supply voltage	5 Vdc ±10 percent	
Clock input		
Frequency	8 MHz maximum (C_L = 40 pF)	
Pulse width (high level)	50 ns minimum	
Trigger edge	Positive	
Input voltage		
Low level	-0.5 V minimum, 0.7 V maximum	
High level	2.0 V minimum, V_{CC} maximum	
Input current (P0, P1, P2, and P3)		
Low level (V_{IL} = 0.0 V)	-0.29 mA maximum	
(V_{IL} = 0.3 V)	-0.27 mA maximum	
High level (V_{IH} = 2.4 V)	13.3 µA maximum	
(V_{IH} = 5.5 V)	1.0 mA maximum	
Input current (CEP-1)		
Low level (V_{IL} = 0.0 V)	-0.43 mA maximum	
(V_{IL} = 0.3 V)	-0.40 mA maximum	
High level (V_{IH} = 2.4 V)	20 µA maximum	
(V_{IH} = 5.5 V)	1 mA maximum	
Input current (CP-2 and CP-3)		
Low level (V_{IL} = 0.0 V)	-1.72 mA maximum	
(V_{IL} = 0.3 V)	-1.6 mA maximum	
High level (V_{IH} = 2.4 V)	80 µA maximum	
(V_{IH} = 5 5 V)	2 mA maximum	
Input current (CP-1, CET, and PE)		
Low level (V_{IL} = 0.0 V)	-0.86 mA maximum	
(V_{IL} = 0.3 V)	-0.8 mA maximum	
High level (V_{IH} = 2.4 V)	40 µA maximum	
(V_{IH} = 5.5 V)	1 mA maximum	
Input current (CEP-2 and CEP-3)		
Low level (V_{IL} = 0.0 V)	-0.86 mA maximum	
(V_{IL} = 0.3 V)	-0.8 mA maximum	
High level (V_{IH} = 2.4 V)	40 µA maximum	
(V_{IH} = 5.5 V)	2 mA maximum	
Input current (MR-1)		
Low level (V_{IL} = 0.0 V)	-1.88 mA maximum	
(V_{IL} = 0.3 V)	-1.75 mA maximum	
High level (V_{IH} = 2.4 V)	-336 µA maximum	
(V_{IH} = 5.5 V)	3 mA maximum	
Output voltage		
Low level	0.0 V minimum, 0.3 V maximum	
High level	2.4 V minimum, V_{CC} maximum	
Output drive current maximum capability		
Low level (V_{OL} = 0.3 V)	3.2 mA	
High level (V_{OH} = 2.4 V)	-0.32 mA	
Propagation delay time (t_{PHL})	C_L = 100 pF	C_L = 40 pF
CP to TC	58 ns maximum	52 ns maximum
CP to Q	68 ns maximum	62 ns maximum
CET to TC	75 ns maximum	68 ns maximum
MR to Q	111 ns maximum	105 ns maximum

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	C_L = 100 pF	C_L = 40 pF
Propagation delay time (t_{PLH})		
CP to Q	65 ns maximum	59 ns maximum
CP to TC	111 ns maximum	105 ns maximum
CET to TC	70 ns maximum	63 ns maximum
Setup time		
CET and CEP	80 ns minimum	
P0, P1, P2, and P3	110 ns minimum	
PE	80 ns minimum	
Hold time		
CET and CEP	0 ns minimum	
P0, P1, P2, and P3	0 ns minimum	
PE	0 ns minimum	
Pulse width (MR)	65 ns minimum	
Recovery time (MR)	50 ns minimum	

6.5 Module function. This module contains three 4-bit synchronous BCD counters with independent inputs and outputs with the exception that circuits 2 and 3 have a common clock (CP) and count enable parallel (CEP) inputs. All counters have a common master reset (MR) input as shown on figure 1.

6 5.1 Input to output relationship. Each counter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 5/

Synchronous inputs before clock pulse							Required condition of asynchronous inputs	Outputs after clock pulse goes high				
CEP	CET	PE	P3	P2	P1	P0	MR	TC	Q3	Q2	Q1	Q0
L	X	H	D3	D2	D1	D0	H	X	NC	NC	NC	NC
X	L	H	D3	D2	D1	D0	H	L	NC	NC	NC	NC
X	X	L	D3	D2	D1	D0	H	L	D3	D2	D1	D0
X	X	L	H	H	H	H	H	L	H	H	H	H
X	X	L	L	L	L	L	H	L	L	L	L	L
H	H	H	X	X	X	X	H	L	L	L	L	H
H	H	H	X	X	X	X	H	L	L	L	H	L
H	H	H	X	X	X	X	H	L	L	L	H	H
H	H	H	X	X	X	X	H	L	L	H	L	L
H	H	H	X	X	X	X	H	L	L	H	L	H
H	H	H	X	X	X	X	H	L	L	H	H	L
H	H	H	X	X	X	X	H	L	L	H	H	H
H	H	H	X	X	X	X	H	L	H	L	L	L
H	H	H	X	X	X	X	H	H	H	L	L	H

1/ NC = No change from previous state.

2/ D0-D3 = Known input and output data (either high level or low level).

3/ X = Unspecified inputs or outputs (either high level or low level).

4/ L = Low-level voltage.

5/ H = High-level voltage.

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6.5.1.1 Functional description. The module contains three BCD decade counters which are fully synchronous with the clock pulse. Each counter contains four master slave flip-flops which are driven, in parallel, by the clock through a clock buffer. The master is inhibited from further change during the low-to-high transition of the clock. Data is transferred from the masters to the slaves and reflected at the outputs after the masters are inhibited. When the clock is high the masters remain inhibited and the master or slave data path remains established. During the clock high-to-low transition, the slave is inhibited from further change followed by the enabling of the masters for acceptance of new data from the counting logic or the parallel entry logic.

6.5.1.2 Control inputs. There are three control inputs parallel enable (PE), count enable parallel (CEP), and count enable trickle (CET). These control inputs select the mode of operation as shown in table III. When the conditions for counting are satisfied, that is CEP, CET, and PE are high, the rising edge of the clock will change the counters to the state of the count sequence.

TABLE III. Mode selection. 1/ - 3/

PE	CEP	CET	Mode
L	L	L	Preset
L	L	H	Preset
L	H	L	Preset
L	H	H	Preset
H	L	L	No change
H	L	H	No change
H	H	L	No change
H	H	H	Count

1/ H = High-level voltage.
2/ L = Low-level voltage.
3/ H = High-level voltage.

6.5.1.3 Counter setting. The counter can be synchronously preset from the four parallel inputs (P0-P3) when PE is low. When the PE and clock are low, each master of the flip-flops is connected to the appropriate parallel input (P0-P3) and the slaves (outputs) are steady in their previous state. When the clock goes high, this information is transferred to the slaves and reflected at the outputs. The PE overrides both count enables presetting the counter low.

6.5.1.4 State of terminal count. Terminal count is high when the counter is at a terminal count (state 9) and the CET is high

6.5.1.5 Overriding provisions. The asynchronous master reset, common to all counters, overrides all other inputs resetting each output low.

6.5.1.6 Special conditions. If PE goes high while the clock is low and the count enables are not active (low) during the low portion of the clock, the low-to-high clock transition will change Q0-Q3 to the P0-P3 data that existed at the setup time before the rising edge of PE.

a. If the count enables are active (high) during some portion of the clock low period and the PE is high during the entire clock low period then the subsequent low-to-high transition will change Q0-Q3 to the next count value.

b If PE goes high while the clock is low and the count enables are active (high) during some portion of the remaining clock low period then the counter will perform a mixture of counting and loading. On the low-to-high clock transition, outputs Q0-Q3 will change as the count sequence or the loading requires. Only the outputs that would not change in the count sequence and are also reloaded with their present value stay constant.

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6.5.1.7 Common CEP and master reset. Two of the counters in the module have a common CEP, and all three have a common master reset.

6.5.1.8 High speed look-ahead technique. Multistage synchronous counting at high speeds is made possible with a high speed look-ahead technique utilizing the count enables and terminal count logic.

6.5.1.9 Terminal count. TC (terminal count) = CET . Q0 . $\overline{Q1}$. $\overline{Q2}$. Q3.

6.5.2 Paralleling outputs. Paralleling outputs is not permissible.

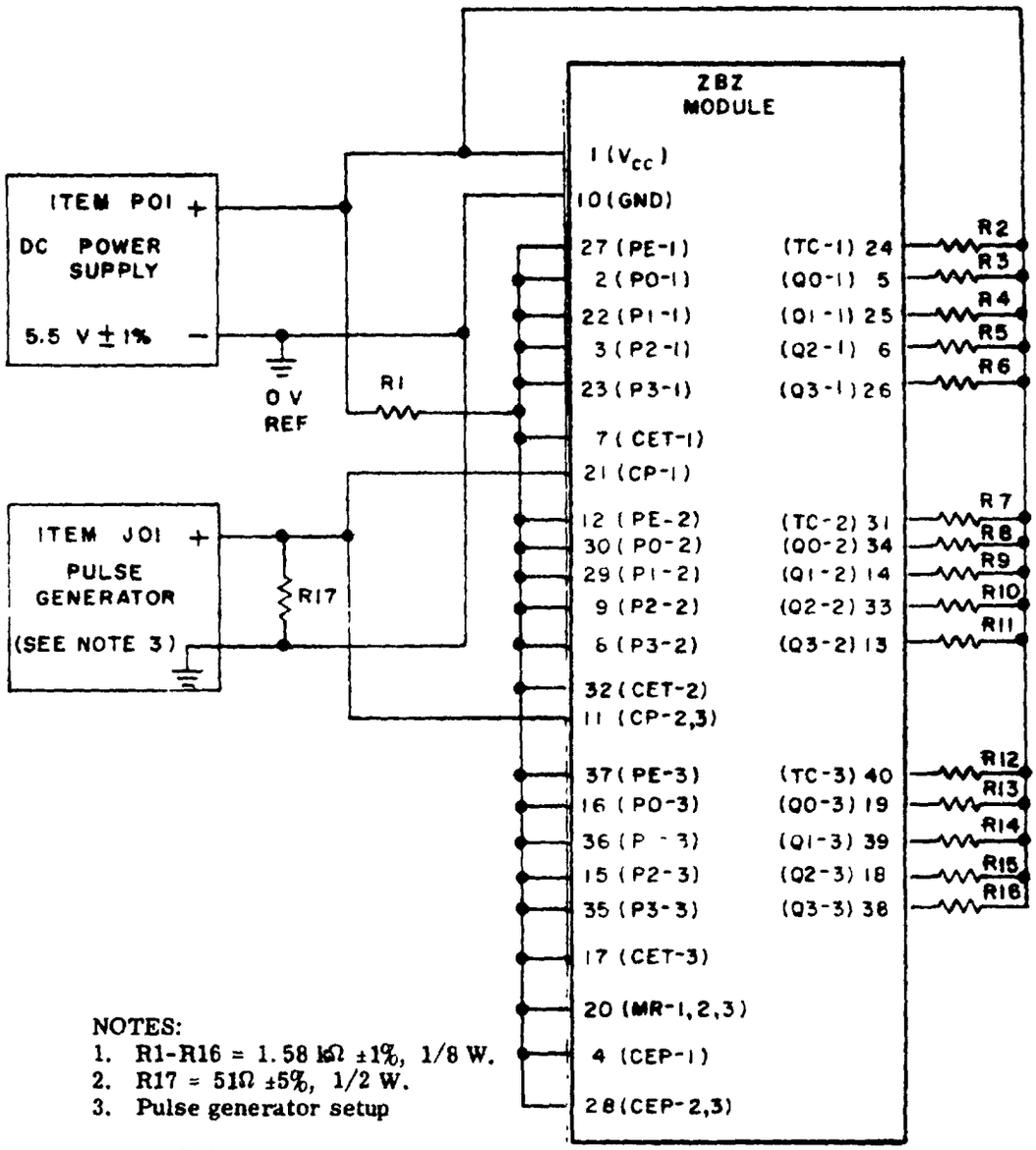
* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
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Preparing activity
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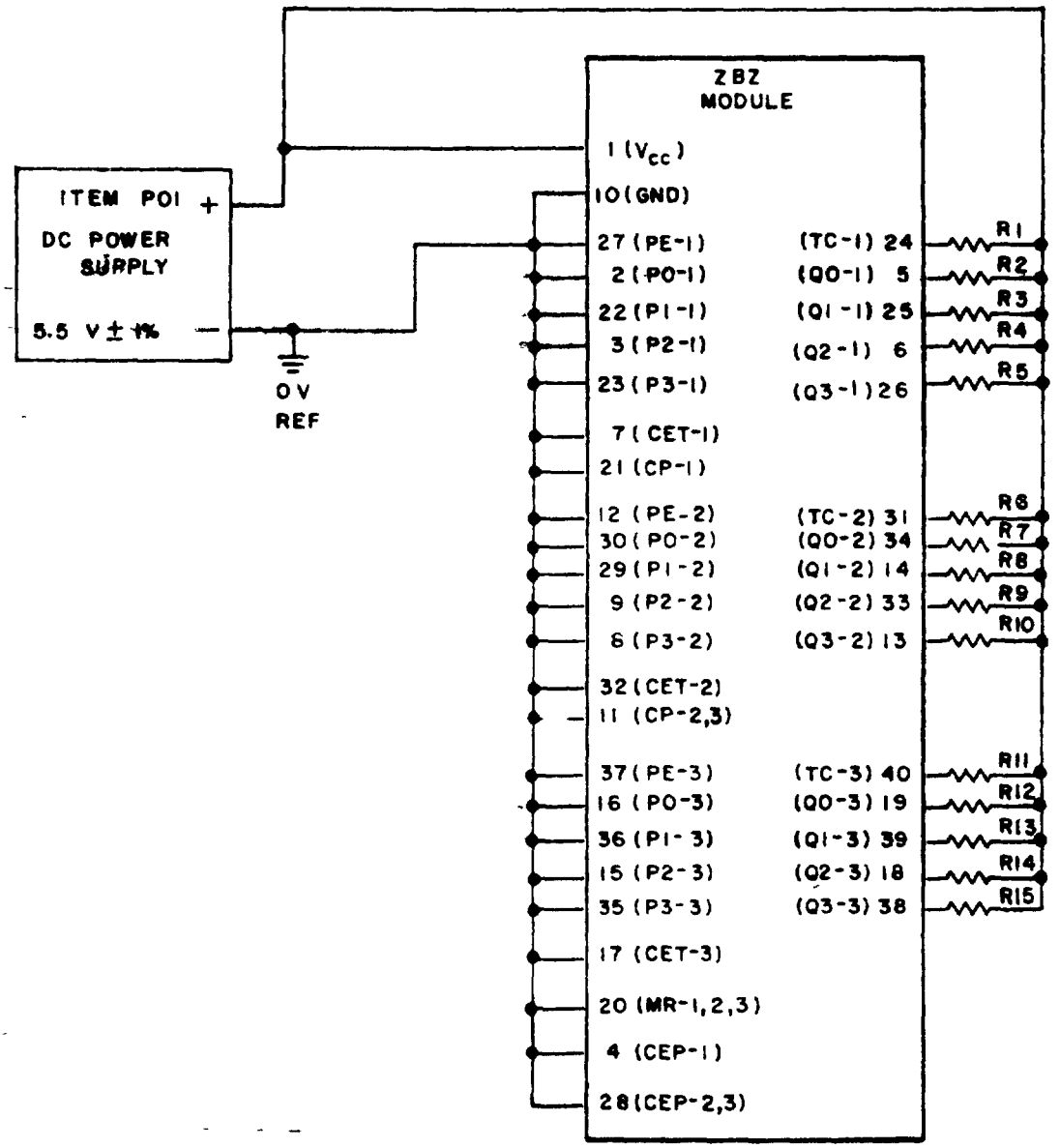
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NOTES:
1. R1-R16 = 1.58 kΩ ±1%, 1/8 W.
2. R17 = 51Ω ±5%, 1/2 W.
3. Pulse generator setup
   a. Pulse width = 1 ±0.1 μs.
   b. Pulse repetition period = 5 ±0.5 μs.
   c. $t_{TLH} = t_{THL}$ = 50 ±25 ns.
   d. Amplitude = 4.0 ±1.0 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R15 = 1.58 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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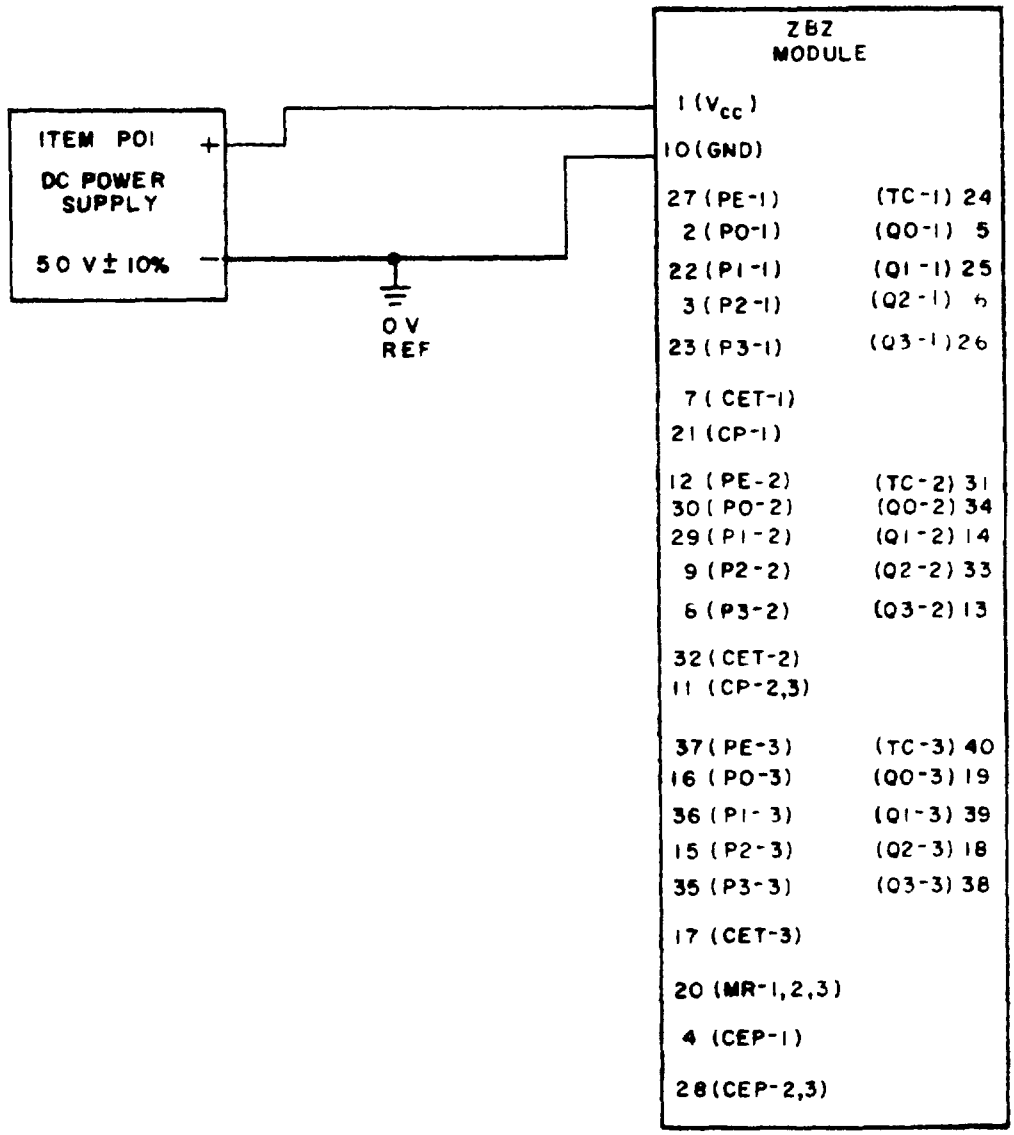


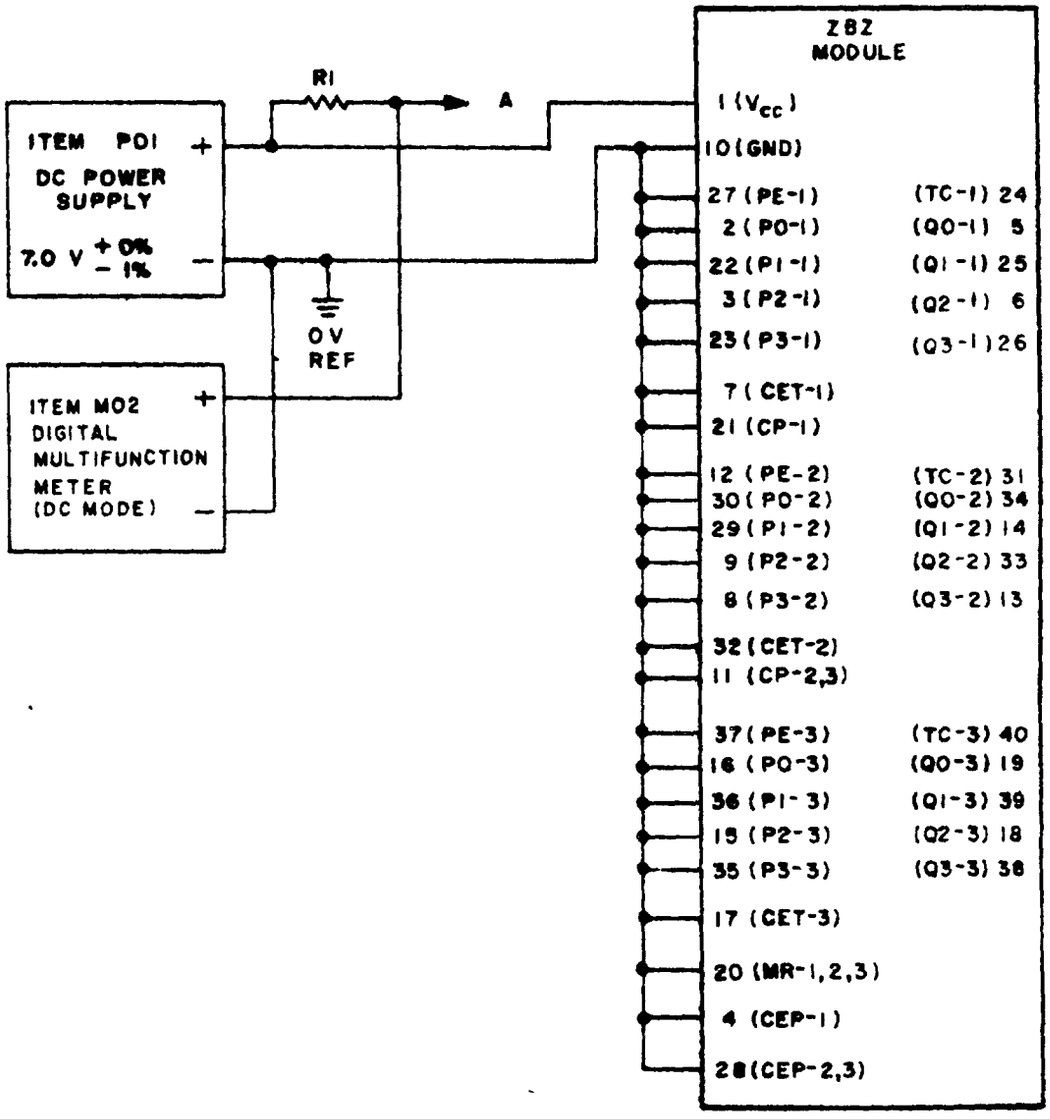
FIGURE 4. Vibration test circuit.

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TABLE IV V_{OUT} (low-loaded) test

Test conditio-	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	O3-1 V_{OUT} (low loaded)	26	0.01	0.60	V	5
A2	Q2-1	6				
A3	C1-1	25				
A4	O0-1	5				
A5	TC-1	24				
A6	Q3-2	13				
A7	O2-2	33				
A8	O1-2	14				
A9	O0-2	34				
A10	TC-2	31				
A11	Q3-3	38				
A12	Q2-3	18				
A13	Q1-3	39				
A14	Q0-3	19				
A15	TC-3	40				

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NOTE: R1 = 1.58 kΩ ±1%, 1/8 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

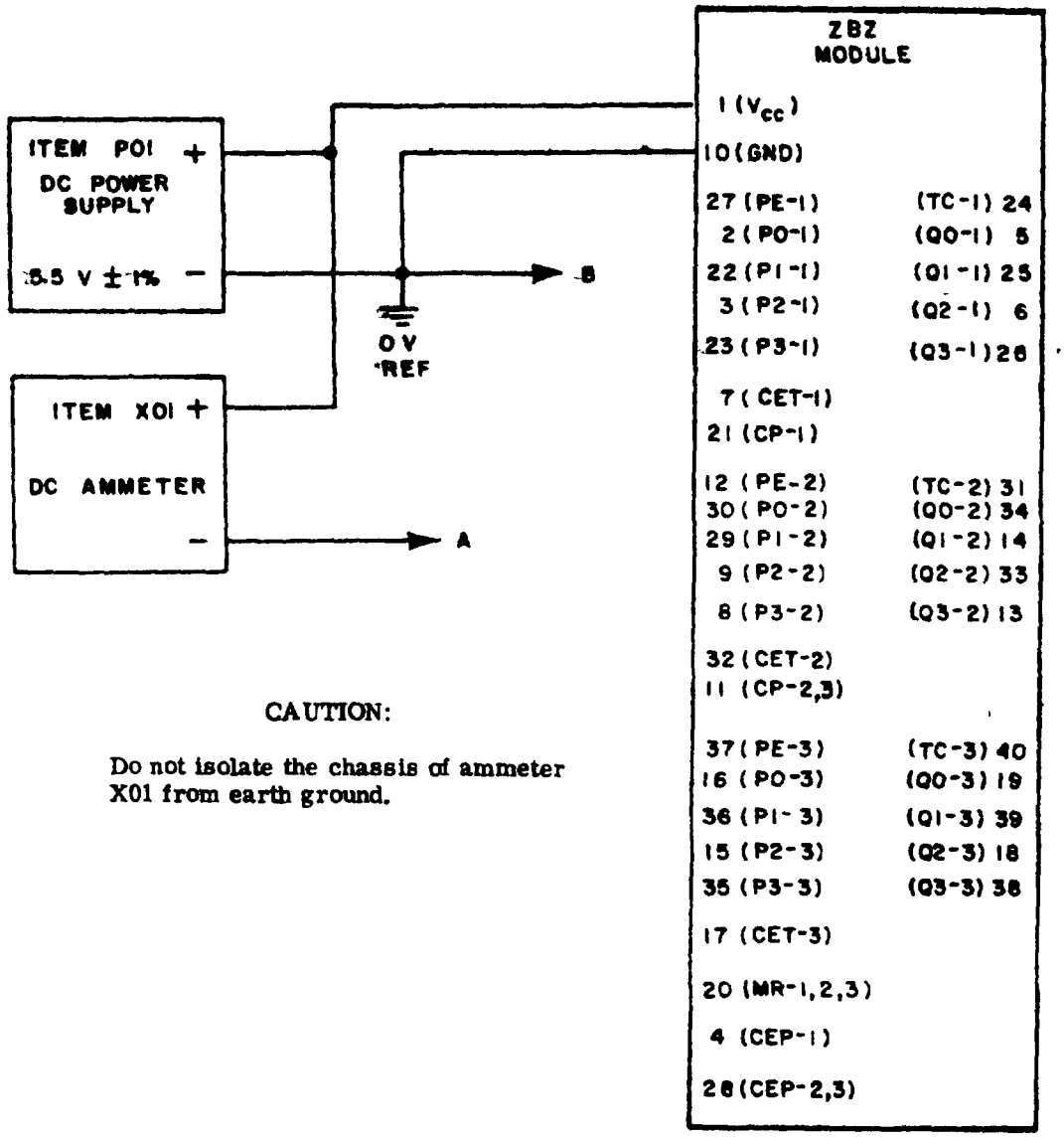
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	CEP-1 Input breakdown current	4	2,3,7,20,21, 22,23,27	0.0	1.0	mA	6
B2	CP-1	21	2,3,4,7,20, 22,23,27				
B3	CET-1	7	2,3,4,20,21, 22,23,27				
B4	PE-1	27	2,3,4,7,20, 21,22,23				
B5	PE-2	12	8,9,11,20,28, 29,30,32,37				
B6	CET-2	32	8,9,11,12,20, 28,29,30,37				
B7	CET-3	17	11,15,16,20, 28,35,36,37				
B8	PO-1	2	3,4,7,20,21, 22,23				
B9	P1-1	22	2,3,4,7,20, 21,23				
B10	P2-1	3	2,4,7,20,21, 22,23				
B11	P3-1	23	2,3,4,7,20, 21,22				
B12	PO-2	30	8,9,11,12,20, 28,29,32				
B13	P1-2	29	8,9,11,20,28, 30,32				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B14	P2-2 Input breakdown current	9	8,11,20,28, 29,30,32	0 0	1.0	mA	6
B15	P3-2	8	9,11,20,28, 29,30,32				
B16	P3-3	35	11,15,16,17, 20,28,36				
B17	P2-3	15	11,16,17,20, 28,35,36				
B18	P1-3	36	11,15,16,17, 20,28,35				
B19	P0-3	16	11,15,17,20, 28,35,36				
B20	PE-3	37	11,15,16,17, 20,28,35,36				
B21	CEP-2,3	28	8,9,11,12,15, 16,17,20,29, 30,32,35,36, 37		2.0		
B22	CP-2,3	11	8,9,12,15,16, 17,20,28,29, 30,32,35,36, 37		2.0		
B23	MR-1,2,3	20	2,3,4,7,8,9, 11,12,15,16, 17,21,22,23, 27,28,29,30, 32,35,36,37		3.0		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	MR-1,2,3 Input clamp diode voltage	20	-0.4	-1.5	V	7
C2	CP-1	21				
C3	PO-1	2				
C4	P1-1	22				
C5	P2-1	3				
C6	P3-1	23				
C7	CET-1	7				
C8	PE-1	27				
C9	CEP-1	4				
C10	CP-2	11				
C11	PO-2	30				
C12	P1-2	29				
C13	P2-2	9				
C14	CET-2	32				
C15	PE-2	12				
C16	P3-2	8				
C17	CET-3	17				
C18	PE-3	37				

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TABLE VI. Input clamp diode voltage test - Continued.

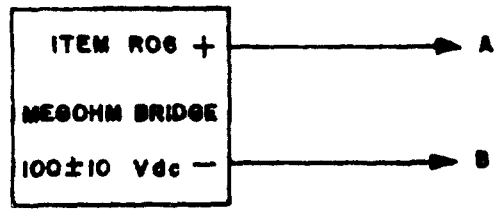
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C19	P3-3 Input clamp diode voltage	35	-0.4	-1.5	V	7
C20	P2-3	15	↓	↓	↓	↓
C21	P1-3	36				
C22	P0-3	16				
C23	CEP-2,3	28				

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TABLE VII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A	B	Min	Max		
D	Isolation	All pins	Frame	10	∞	MΩ	8

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

Z8Z MODULE	
1 (V_{cc})	
10 (GND)	
27 (PE-1)	(TC-1) 24
2 (PO-1)	(QO-1) 5
22 (P1-1)	(Q1-1) 25
3 (P2-1)	(Q2-1) 6
23 (P3-1)	(Q3-1) 26
7 (CET-1)	
21 (CP-1)	
12 (PE-2)	(TC-2) 31
30 (PO-2)	(QO-2) 34
29 (P1-2)	(Q1-2) 14
9 (P2-2)	(Q2-2) 33
8 (P3-2)	(Q3-2) 13
32 (CET-2)	
11 (CP-2,3)	
37 (PE-3)	(TC-3) 40
16 (PO-3)	(QO-3) 19
36 (P1-3)	(Q1-3) 39
15 (P2-3)	(Q2-3) 18
35 (P3-3)	(Q3-3) 38
17 (CET-3)	
20 (MR-1,2,3)	
4 (CEP-1)	
28 (CEP-2,3)	

FIGURE 8. Isolation test circuit for table VII.

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TABLE VIII. Functional, static output voltage, and input threshold check test

Test condition	Parameter	Connect								Output logic states					Initial limits 25° ±5°C ambient		Unit	Test figure
		A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
E1	Functional count, high level	Open	2,3,22, 23,G	4,7,27	Open	Open	21	B	20	See table IX					See figure 9, note 6			9
E2	Functional count, high level	Open	8,9,29, 30,G	12,28, 32	Open		11			See table IX								
E3	Functional count, high level	Open	15,16, 35,36,G	17,28, 37	Open		11			See table IX								
F1	Functional CET-1, low level	2,4, 23,27	3,22,G	Open	7		21			L	L	L	L	L				
F2	Functional PE-1, low level	2,4,23	3,22,G		7,27		21			L	H	L	L	H				
F3	Functional CET-2, low level	8,12, 28,30	9,29,G		32		11			L	L	L	L	L				
F4	Functional PE-2, low level	8,30, 28	9,29,G		12,32					L	H	L	L	H				
F5	Functional CET-3, low level	16,28, 35,37	15,36,G		17					L	L	L	L	L				
F6	Functional PE-3, low level	16,28, 35	15,36,G		17,37					L	H	L	L	H				
F7	Functional CEP-1, low level	2,3,7, 22,23, 27	G		4		21			L	L	L	L	L				
F8	Functional CEP-2,3, low level	8,9, 12,29, 30,32	G		28		11			L	L	L	L	L				
F9	Functional CEP-2,3, low level	15,16, 17,35, 36,37	G		28		11			L	L	L	L	L				

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TABLE VIII. Functional, static output voltage, and input threshold check test - Continued.

Test condition	Parameter	Connect								Output logic states					Initial limits 25° ±5°C ambient		Unit	Test figure
		A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
G1	TC-1 Static output voltage, high level	4,7	27,G	2,23	3,22	24	21	B	20						2.4	4.5	V	9
G2	Q3-1	Open	4,7,27, G	23	2,3,22	26												
G3	Q2-1			3	2,22, 23	6												
G4	Q1-1			22	2,3,23	25												
G5	Q0 1			2	3,22, 23	5												
G6	TC-2	28,32	12,G	8,30	9,29	31	11											
G7	Q3-2	Open	12,28, 32,G	8	9,29, 30	13												
G8	Q2-2			9	8,29, 30	33												
G9	Q1-2			29	8,9,30	14												
G10	Q0-2			30	8,9,29	34												
G11	TC-3	28,17	37,3	16,35	15,36	40												
G12	Q3-3	Open	17,28, 37,G	35	15,16, 36	38												
G13	Q2-3	Open	17,28, 37,G	15	16,35, 36	18												

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TABLE VIII. Functional, static output voltage, and input threshold check test - Continued

Test condition	Parameter	Connect								Output logic states					Initial limits 25° ±5°C ambient		Unit	Test figure
		A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
G14	Q1-3 Static output voltage, high level	Open	17,28, 37,G	36	15,16, 35	39	11	B	20						2.4	4.5	V	9
G15	Q0-3 Static output voltage, high level		17,28, 37,G	16	15,35, 36	19	11								2 4	4 5		
H1	Q3-1 Static output voltage, low level		4,7,27, G	2,3,22	23	26	21								0.00	0 30		
H2	Q2-1			2,22, 23	3	6												
H3	Q1-1			2,3,23	22	25												
H4	Q0-1			3,24, 23	2	5												
H5	Q3-2		12,28, 32,G	9,29, 30	8	13	11											
H6	Q2-2			8,29, 30	9	33												
H7	Q1-2			8,9,30	29	14												
H8	Q0-2			8,9,29	30	34												
H9	Q3-3		17,28, 37,G	15,16, 36	35	38												
H10	Q2-3		17,28, 37, G	16,35, 36	15	18												

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TABLE VIII Functional, static output voltage, and input threshold check test - Continued

Test condition	Parameter	Connect								Output logic states					Initial limits 25° ±5°C ambient		Unit	Test figure
		A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
H11	Q1-3 Static output voltage, low level	Open	17,28, 37,G	15,16, 35	36	39	11	B	20						0.00	0.30	V	9
H12	Q0-3		17,28, 37,G	15,35, 36	16	19	11											
H13	TC-1		4,7,27, G	2,3,23	22	24	21											
H14	TC-2		12,28, 32,G	8,9,30	29	31	11											
H15	TC-'		17,28, 37,G	15,35, 36	16	40	11											
J1	MR-1,2,3 Input threshold check, low level	2,3,4, 7,22, 23	27	Open	G	Open	20	D	21	L	L	L	L	L	See figure 9, note 6			
J2	MR-1,2,3 Input threshold check, low level	8,9, 28,29, 30,32	12	Open	G		20	D	11	L	L	L	L	L				
J3	MR-1,2,3 Input threshold check, low level	15,16, 17,28, 35,36	37	Open	G		20	D	11	L	L	L	L	L				
K1	CET-1 Input threshold check, high level	2,4, 23	3,22, 27,G	7	Open		21	B	20	H	H	L	L	H				
K2	CET-2 Input threshold check, high level	8,28, 30	9,12, 29,G	32	Open		11	B	20	H	H	L	L	H				
K3	CET-3 Input threshold check, high level	16,28, 35	15,36, 37,G	17	Open		11	B	20	H	H	L	L	H				

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TABLE VIII. Functional, static output voltage, and input threshold check test - Continued.

Test condition	Parameter	Connect								Output logic states					Initial limits 25° ±5°C ambient		Unit	Test figure
		A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
K4	CEP-1 Input threshold check, high level	2,7, 23,27	3,22,G	4	Open	Open	21	B	20	L	L	L	L	H	See figure 9, note 6			9
K5	CEP-2,3 Input threshold check, high level	8,30, 32,29	9,12,G	28	Open	Open	11	B	20	L	L	L	L	H	See figure 9, note 6			9
K6	CEP-2,3 Input threshold check, high level	16,17, 35,37	15,36,G	28	Open	Open	11	B	20	L	L	L	L	H	See figure 9, note 6			9

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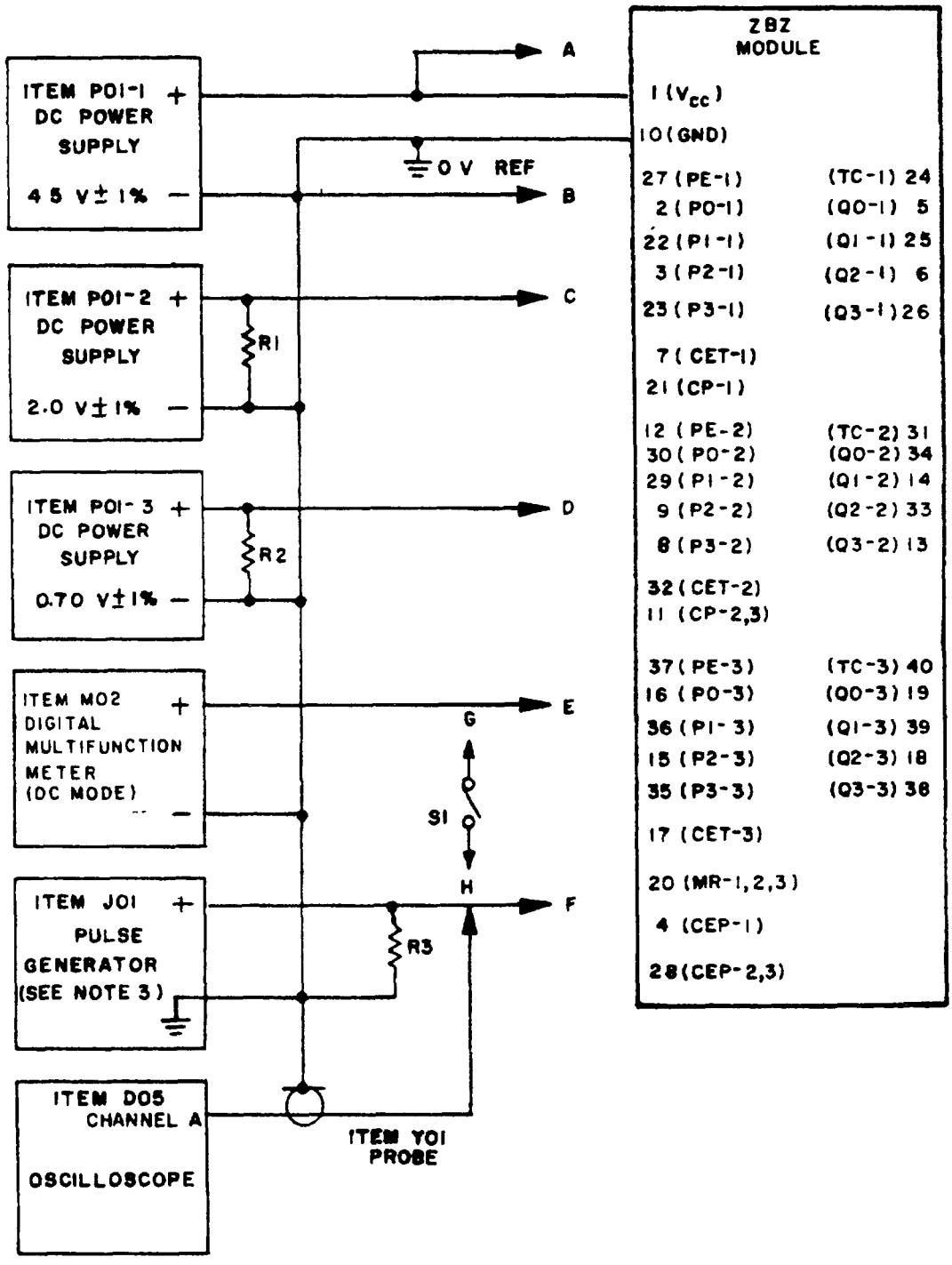
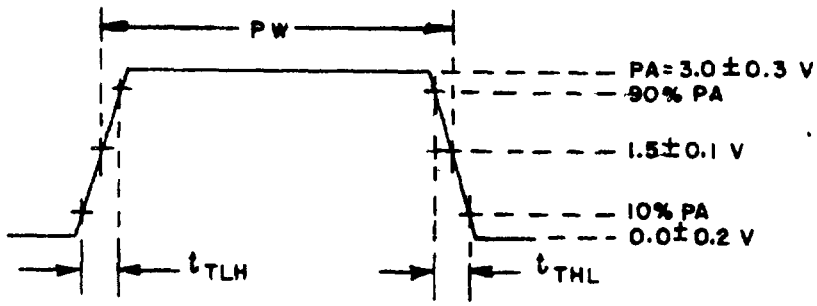


FIGURE 9. Functional, static output voltage, and input threshold check test circuit for table VIII.

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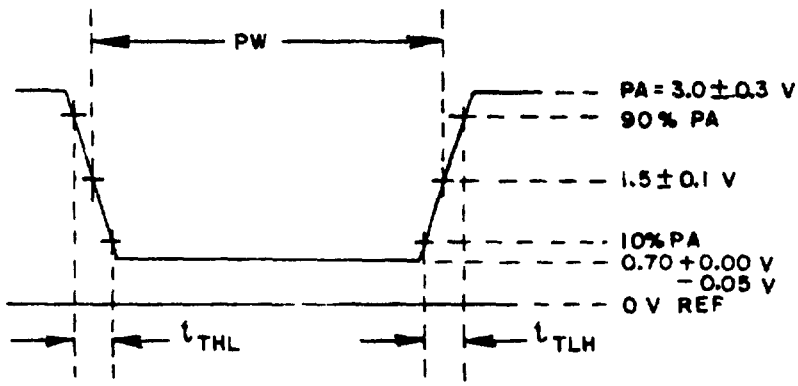
NOTES.
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. S1 = SPST.
3. Input pulse parameters:

a. For test conditions E, F, G, H, and K.



(1) PW = 500 ±100 ns.
(2) $t_{TLH} = t_{THL}$ = 50 ±25 ns.
(3) PRR = 30 kHz ±50%.

b. For test condition J.



(1) PW = 500 ±100 ns.
(2) $t_{TLH} = t_{THL}$ = 20 ±2 ns.
(3) PRR = 30 kHz ±50%.

c. The pulse width, transition times, and pulse amplitude shall be adjusted by using oscilloscope D05.

FIGURE 9. Functional, static output voltage, and input threshold check test circuit for table VIII - Continued.

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4. Test procedures:

   a. For test condition E:

      After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With the pulse generator in the single pulse mode, apply 10 pulses to the circuit under test. The output states for each input pulse shall be as specified in table IX.

   b. For test conditions F, G, and H:

      After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With the pulse generator in the single pulse mode, apply one pulse to the circuit under test. The output states and voltage levels shall be as specified in table VIII.

   c. For test condition J.

      Initially switch S1 shall be closed. After making all circuit connections for a test condition, open switch S1 for a minimum of 100 ns, then close switch S1. With the pulse generator in the single pulse mode, apply one pulse to the circuit under test. The output states shall be as specified in table VIII.

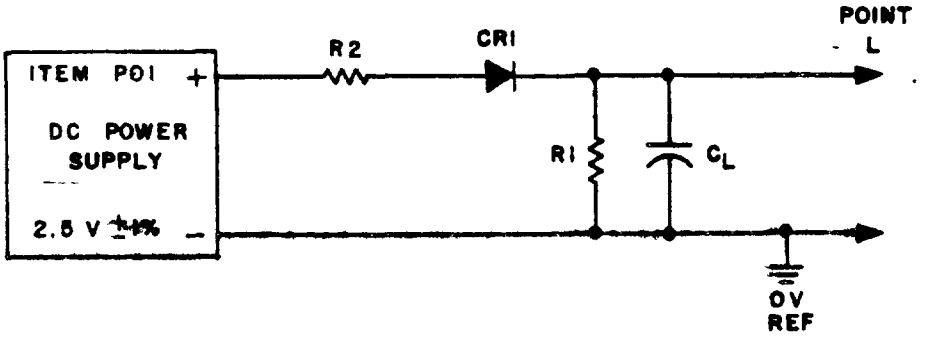
   d. For test condition K.

      After making all connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With the pulse generator in the single pulse mode, apply one pulse to the circuit under test. The output states shall be as specified in table VIII.

5. Load all outputs with a logic load circuit as specified on figure 10.
6. For test conditions E, F, J, and K, verification of the output logic states (L = 0.0 to 0.3 V and H = 2.4 to 4.5 V) is obtained by connecting voltmeter M02 to each output in succession, after input conditions are accomplished.

FIGURE 9. Functional, static output voltage, and input threshold check test circuit for table VIII - Continued.

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NOTES.
1. CR1 = 1N4150, or equivalent.
2. C_L = 40 ±5 pF (including probe and parasitic capacitance) for test conditions G, H, S, U, W, Y, AA, CC, EE, FF, and GG.
3. C_L = 100 ±5 pF (including probe and parasitic capacitance) on the output under test for test conditions T, V, X, Z, BB, and DD.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions HH and JJ.
5. R1 = Select a value for a current flow of 0.32 mA ±1% into point L when point L is held at 2.4 V.
6. R2 = Select a value for a current flow of 3.2 mA ±1% when point L is held at 0.3 V.

FIGURE 10. ZBZ logic load circuit.

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TABLE IX. Functional output logic states. 1/, 2/

Test No.	Total pulses	Q0	Q1	Q2	Q3	TC
Initial state	0	L	L	L	L	L
1	1	H	L	L	L	L
2	2	L	H	L	L	L
3	3	H	H	L	L	L
4	4	L	L	H	L	L
5	5	H	L	H	L	L
6	6	L	H	H	L	L
7	7	H	H	H	L	L
8	8	L	L	L	H	L
9	9	H	L	L	H	H
10	10	L	L	L	L	L

1/ L = Low-level voltage.
2/ H = High-level voltage.

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TABLE X. Input current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	MR-1,2,3 Input current, low level	Open	Open	20	Open	0.82	1.88	mA	11
L2	CP-1			21		0.10	0.86		
L3	CET-1			7					
L4	PE-1			27					
L5	CEP-2,3			28					
L6	CET-2			32					
L7	CET-3			17					
L8	PE-2			12					
L9	PE-3			37					
L10	CP-2,3			11			1.72		
L11	CEP-1			4			0.43		
L12	PO-1			2	27		0.29		
L13	P1-1			22					
L14	P2-1			3					
L15	P3-1			23					
L16	PO-2			30	12				
L17	P1-2			29	12				

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TABLE X. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L18	P2-2 Input current, low level	Open	Open	9	12	0.10	0.29	mA	11
L19	P3-2			8	12				
L20	P3-3			35	37				
L21	P2-3			15					
L22	P1-3			36					
L23	P0-3			16					
M1	MR-1,2,3 Input current, high level	20		Open	2,3,4, 7,8,9, 11,12, 15,16, 17,21, 22,23, 27,28, 29,30, 32,35, 36,37	-0.228	-0.336		
M2	CP-1	21			2,3,4, 7,20, 22,23, 27	0	40	µA	
M3	CET-1	7			2,3,4, 20,21, 22,23, 27	0	40	µA	
M4	PE-1	27			2,3,4, 7,20, 21,22, 23	0	40	µA	

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TABLE X. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M5	CEP-2,3 Input current, high level	28	Open	Open	8,9, 11,12, 15,16, 17,20, 29,30, 32,35, 36,37	0	40	µA	11
M6	CET-2	32			8,9, 11,12, 20,28, 29,30				
M7	CET-3	17			11,15, 16,20, 28,35, 36,37				
M8	PE-2	12			8,9, 11,20, 28,29, 30,32				
M9	PE-3	37			11,15, 16,17, 20,28, 35,36				
M10	CP-2,3	11			8,9, 12,15, 16,17, 20,28, 29,30, 32,35, 36,37		80		

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TABLE X. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M11	CEP-1 Input current, high level	4	Open	Open	2,3,7, 20,21, 22,23, 27	0	20	μA	11
M12	PO-1	2			3,4,7, 20,21, 22,23, 27	0.0	13.3		
M13	P1-1	22			2,3,4, 7,20, 21,23, 27				
M14	P2-1	3			2,4,7, 20,21, 22,23, 27				
M15	P3-1	23			2,3,4, 7,20, 21,22, 27				
M16	PO-2	30			8,9, 11,12, 20,28, 29,32				
M17	P1-2	29			8,9, 11,12, 20,28, 30,32				

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TABLE X Input current (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M18	P2-2	Input current, high level	9	Open	Open	8,11, 12,20, 28,29, 30,32	0.0	13.3	μA	11
M19	P3-2		8			9,11, 12,20, 28,29, 30,32				
M20	P3-3		35			11,15, 16,17, 20,28, 36,37				
M21	P2-3		15			11,16, 17,20, 28,35, 36,37				
M22	P1-3		36			11,15, 16,17, 20,28, 35,37				
M23	P0-3		16			11,15, 17,20, 28,35, 36,37				

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TABLE XI. Output short circuit current test.

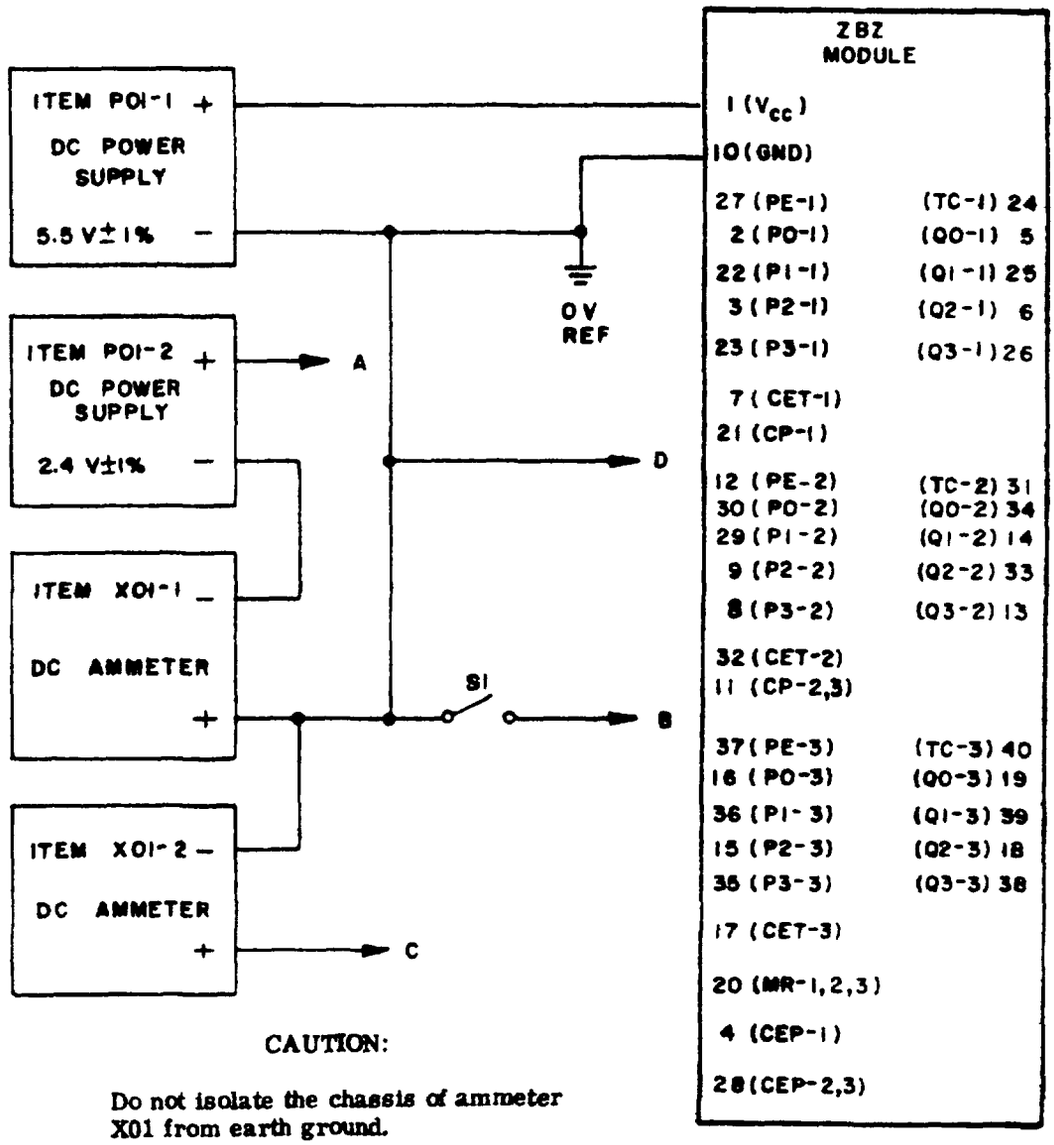
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Q3-1 Output short circuit current	Open	11,21	26	12,27, 37	2.5	25	mA	11
N2	Q2-1			6					
N3	Q1-1			25					
N4	Q0-1			5					
N5	TC-1			24	3,12, 22,27, 37				
N6	Q3-2			13	12,27, 37				
N7	Q2-2			33					
N8	Q1-2			14					
N9	Q0-2			34					
N10	TC-2			31	9,12, 27,29, 37				
N11	Q3-3			18	12,27, 37				
N12	Q2-3			38	12,27, 37				
N13	Q1-3			39	12,27, 37				

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TABLE XI. Output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N14	QO-3 Output short circuit current	Open	11,21	19	12,27, 37	2.5	25	mA	11
N15	TC-3 Output short circuit current	Open	11,21	40	12,15, 27,36, 37	2.5	25	mA	11

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPST.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required
3. After making all circuit connections for a test condition except point C, close switch S1 for a minimum of 100 ns, then open switch S1. Then connect point C, and take measurement.
4. Only one output shall be shorted at a time, and the measurement taken within 10 seconds.

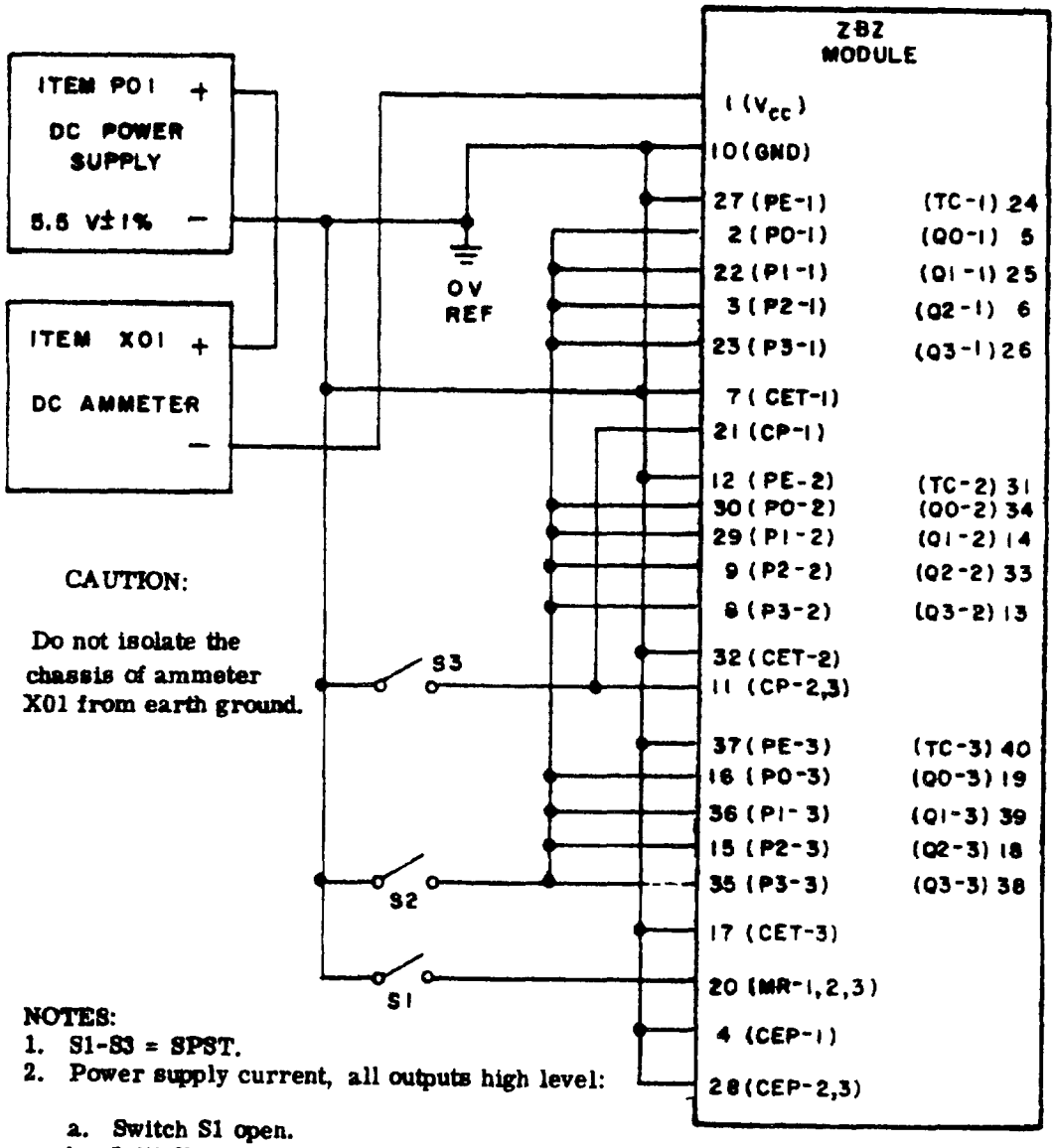
FIGURE 11. Input current (high-level and low-level) and output short circuit current test circuit for tables X and XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
P1	Power supply current, high level	1.0	82.5	mA	12
P2	Power supply current, low level	1.0	82.5	mA	12

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NOTES:
1. S1-S3 = SPST.
2. Power supply current, all outputs high level:
   a. Switch S1 open.
   b. Initially switch S2 shall be open and switch S3 closed.
   c. Open switch S3 for 100 ns minimum, then close switches S3 and S2.
   d. Take measurement.
3. Power supply current; all outputs low level:
   a. Close switches S1, S2, and S3.
   b. Take measurement.

FIGURE 12. Power supply current test circuit for table XII.

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TABLE XIII. Input capacitance test.

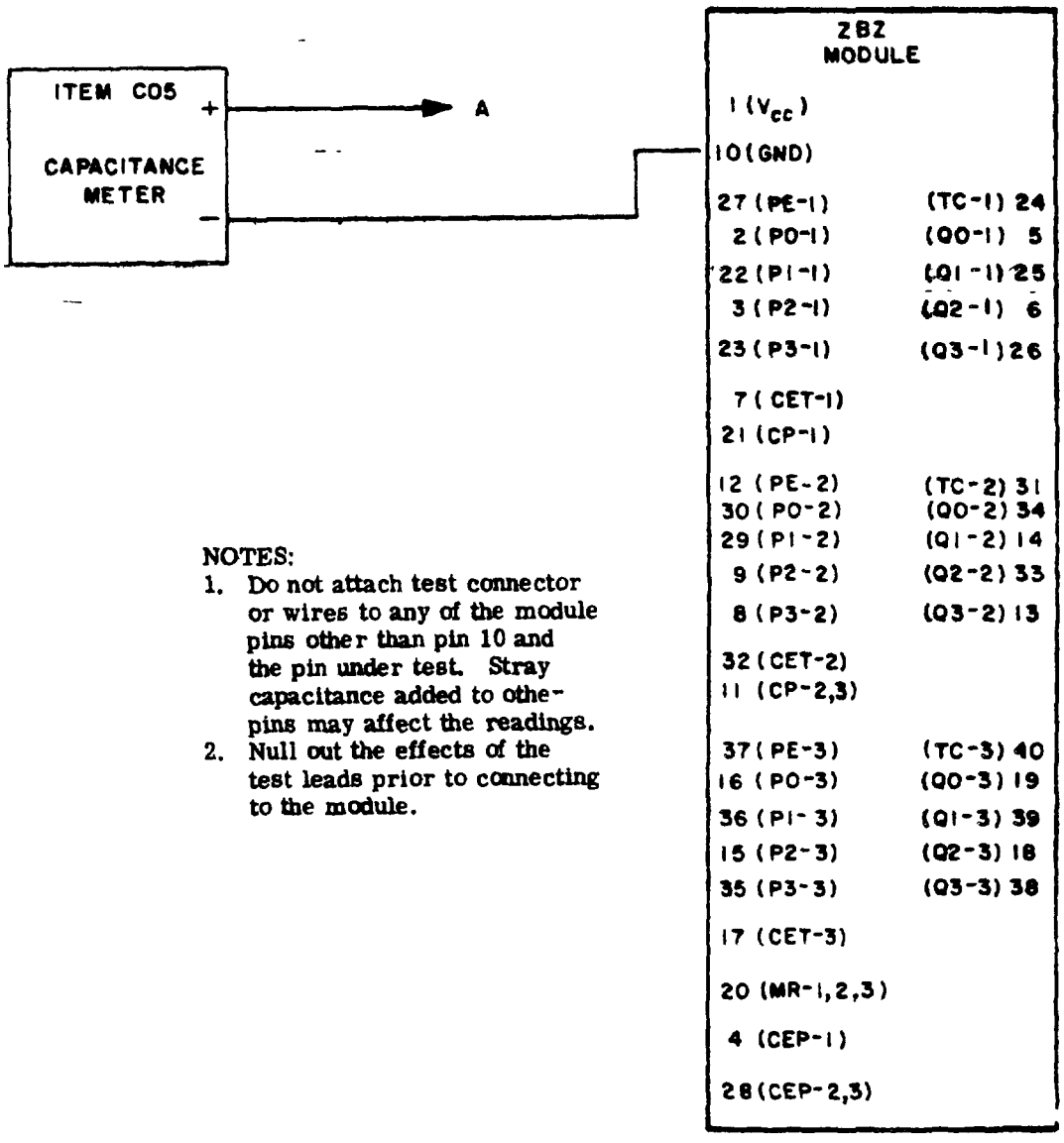
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	MR-1,2,3 Input capacitance	20	3	45	pF	13
Q2	CEP-2,3	28	2	30		
Q3	CP-2,3	11		30		
Q4	PE-1	27		20		
Q5	CP-1	21				
Q6	PO-1	2				
Q7	P1-1	22				
Q8	P2-1	3				
Q9	P3-1	23				
Q10	CET-1	7				
Q11	CEP-1	4				
Q12	PE-2	12				
Q13	PO-2	30				
Q14	P1-2	29				
Q15	P2-2	9				
Q16	P3-2	8				
Q17	CET-2	32				
Q18	PO-3	16				

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TABLE XIII. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q19	P1-3 Input capacitance	36	2	20	pF	13
Q20	P2-3	15	↓	↓	↓	↓
Q21	P3-3	35				
Q22	CET-3	17				
Q23	PE-3	37				

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to othe- pins may affect the readings.
2. Null out the effects of the test leads prior to connecting to the module.

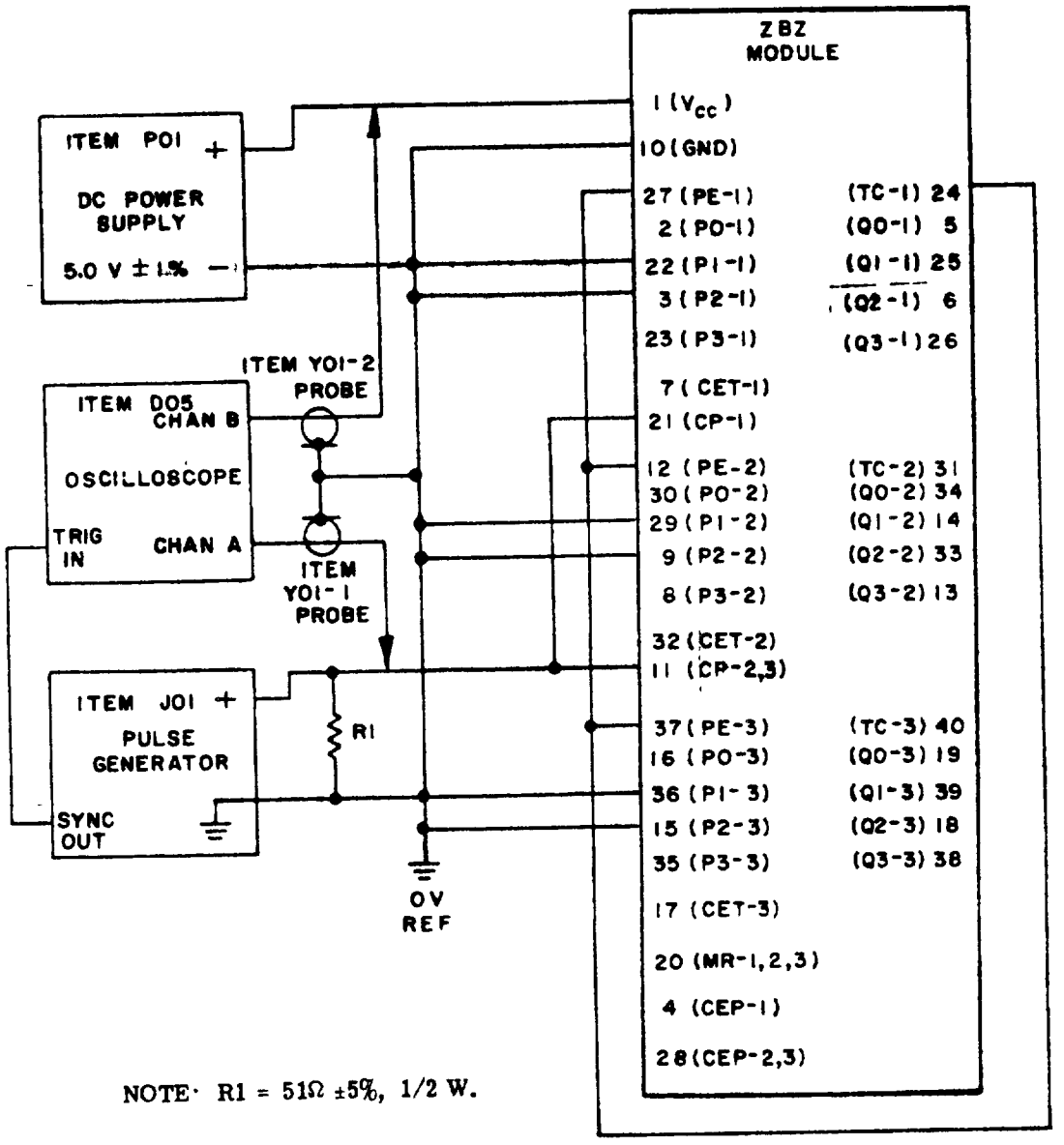
FIGURE 13. Input capacitance test circuit for table XIII.

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TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	14,15

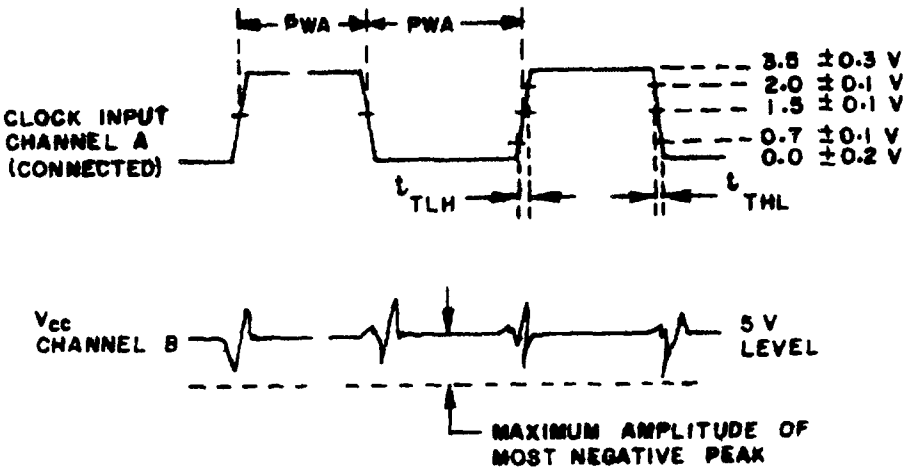
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 14. Power supply transient amplitude test circuit for table XIV.

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NOTE: Input pulse parameters:

a. PWA = 500 ±50 ns.
b. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).

FIGURE 15 Power supply transient amplitude test circuit waveforms for table XIV

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TABLE XV. Propagation delay time (t_{PHL}) test

Test condition	Parameter	Connect				Initial limits 25° + 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
S1	CP-1 to TC-1 Propagation delay time (t_{PHL}), C_L = 40 pF	21	3,22	24	26 to 27	10	49	ns	16,17
S2	CP-2,3 to TC-2	11	9,29	31	12 to 13	10	49		
S3	CP-2,3 to TC-3	11	15,36	40	37 to 38	10	49		
S4	CP-1 to Q0-1	21	3,22	5	26 to 27	15	59		
S5	CP-1 to Q3-1		3,22	26	5 to 27				
S6	CP-1 to Q1-1		23	25	6 to 27				
S7	CP-1 to Q2-1		23	6	25 to 27				
S8	CP-2,3 to Q0-2	11	9,29	34	12 to 13				
S9	CP-2,3 to Q3-2		9,29	13	12 to 34				
S10	CP-2,3 to Q1-2		8	14	12 to 33				
S11	CP-2,3 to Q2-2		8	33	12 to 14				
S12	CP-2,3 to Q0-3		15,36	19	37 to 38				
S13	CP-2,3 to Q3-3		15,36	38	19 to 37				
S14	CP-2,3 to Q1-3		35	39	18 to 37				
S15	CP-2,3 to Q2-3		35	18	37 to 39				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
T1	CP-1 to TC-1 Propagation delay time (t_{PHL}), C_L = 100 pF	21	3,22	24	26 to 27	10	55	ns	16,17
T2	CP-2,3 to TC-2	11	9,29	31	12 to 13	10	55		
T3	CP-2,3 to TC-3	11	15,36	40	37 to 38	10	55		
T4	CP-1 to Q0-1	21	3,22	5	26 to 27	15	65		
T5	CP-1 to Q3-1		3,22	26	5 to 27				
T6	CP-1 to Q1-1		23	25	6 to 27				
T7	CP-1 to Q2-1		23	6	25 to 27				
T8	CP-2,3 to Q0-2	11	9,29	34	12 to 13				
T9	CP-2,3 to Q3-2		9,29	13	12 to 34				
T10	CP-2,3 to Q1-2		8	14	12 to 33				
T11	CP-2,3 to Q2-2		8	33	12 to 14				
T12	CP-2,3 to Q0-3		15,36	19	37 to 38				
T13	CP-2,3 to Q3-3		15,36	38	19 to 37				
T14	CP-2,3 to Q1-3		35	39	18 to 37				
T15	CP-2,3 to Q2-3		35	18	37 to 39				

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TABLE XVI. Transition time (t_{THL}) test

Test condition	Parameter	Connect				Initial limits 25° + 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
U1	TC-1 Transition time (t_{THL}), C_L = 40 pF	21	3,22	24	26 to 27	2	20	ns	16,17
U2	TC-2	11	9,29	31	12 to 13				
U3	TC-3	11	15,36	40	37 to 38				
U4	QO-1	21	3,22	5	26 to 27				
U5	Q3-1		3,22	26	5 to 27				
U6	Q1-1		23	25	6 to 27				
U7	Q2-1		23	6	25 to 27				
U8	QO-2	11	9,29	34	12 to 13				
U9	Q3-2		9,29	13	12 to 34				
U10	Q1-2		8	14	12 to 33				
U11	Q2-2		8	33	12 to 14				
U12	QO-3		15,36	19	37 to 38				
U13	Q3-3		15,36	38	19 to 37				
U14	Q1-3		35	39	18 to 37				
U15	Q2-3		35	18	37 to 39				

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TABLE XVI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° + 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
V1	TC-1 Transition time (t_{THL}), C_L = 100 pF	21	3,22	24	26 to 27	2	25	ns	16,17
V2	TC-2	11	9,29	31	12 to 13				
V3	TC-3	11	15,36	40	37 to 38				
V4	Q0-1	21	3,22	5	26 to 27				
V5	Q3-1		3,22	26	5 to 27				
V6	Q1-1		23	25	6 to 27				
V7	Q_4-1		23	6	25 to 27				
V8	Q0-2	11	9,29	34	12 to 13				
V9	Q3-2		9,29	13	12 to 34				
V10	Q1-2		8	14	12 to 33				
V11	Q2-2		8	33	12 to 14				
V12	Q0-3		15,36	19	37 to 38				
V13	Q3-3		15,36	38	19 to 37				
V14	Q1-3		35	39	18 to 37				
V15	Q2-3		35	18	37 to 39				

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TABLE XVII Propagation delay time (t_{PLH}) test

Test condition	Parameter	Connect				Initial limits 25° ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
W1	CP-1 to TC-1 Propagation delay time (t_{PLH}), C_L = 40 pF	21	2,3,22	24	26 to 27	20	99	ns	16,18
W2	CP-2,3 to TC-2	11	9,29,30	31	12 to 13	20	99		
W3	CP-2,3 to TC-3	11	15,16,36	40	37 to 38	20	99		
W4	CP-1 to Q0-1	21	2,3,22	5	26 to 27	10	54		
W5	CP-1 to Q3-1		23	26	5 to 27				16,17
W6	CP-1 to Q1-1		3,22,23	25	6 to 27				16,17
W7	CP-1 to Q2-1		3,23	6	25 to 27				16,17
W8	CP-2,3 to Q0-2	11	9,29,30	34	12 to 13				16,18
W9	CP-2,3 to Q3-2		8	13	12 to 34				16,17
W10	CP-2,3 to Q1-2		8,9,29	14	12 to 33				16,17
W11	CP-2,3 to Q2-2		8,9	33	12 to 14				16,17
W12	CP-2,3 to Q0-3		15,16,36	19	37 to 38				16,18
W13	CP-2,3 to Q3-3		35	38	19 to 37				16,17
W14	CP-2,3 to Q1-3		15,35,36	39	18 to 37				16,17
W15	CP-2,3 to Q2-3		15,35	18	37 to 39				16,17

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TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° + 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
X1	CP-1 to TC-1 Propagation delay time (t_{PLH}), C_L = 100 pF	21	2,3,22	24	26 to 27	25	105	ns	16,18
X2	CP-2,3 to TC-2	11	9,29,30	31	12 to 13	25	105		
X3	CP-2,3 to TC-3	11	15,16,36	40	37 to 38	25	105		
X4	CP-1 to QO-1	21	2,3,22	5	26 to 27	15	60		
X5	CP-1 to Q3-1		23	26	5 to 27				16,17
X6	CP-1 to Q1-1		3,22,23	25	6 to 27				16,17
X7	CP-1 to Q2-1		3,23	6	25 to 27				16,17
X8	CP-2,3 to QO-2	11	9,29,30	34	12 to 13				16,18
X9	CP-2,3 to Q3-2		8	13	12 to 34				16,17
X10	CP-2,3 to Q1-2		8,9,29	14	12 to 33				16,17
X11	CP-2,3 to Q2-2		8,9	33	12 to 14				16,17
X12	CP-2,3 to QO-3		15,16,36	19	37 to 38				16,18
X13	CP-2,3 to Q3-3		35	38	19 to 37				16,17
X14	CP-2,3 to Q1-3		19,35,36	39	18 to 37				16,17
X15	CP-2,3 to Q2-3		15,35	18	37 to 39				16,17

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TABLE XVIII. Transition time (t_{TLH}) test.

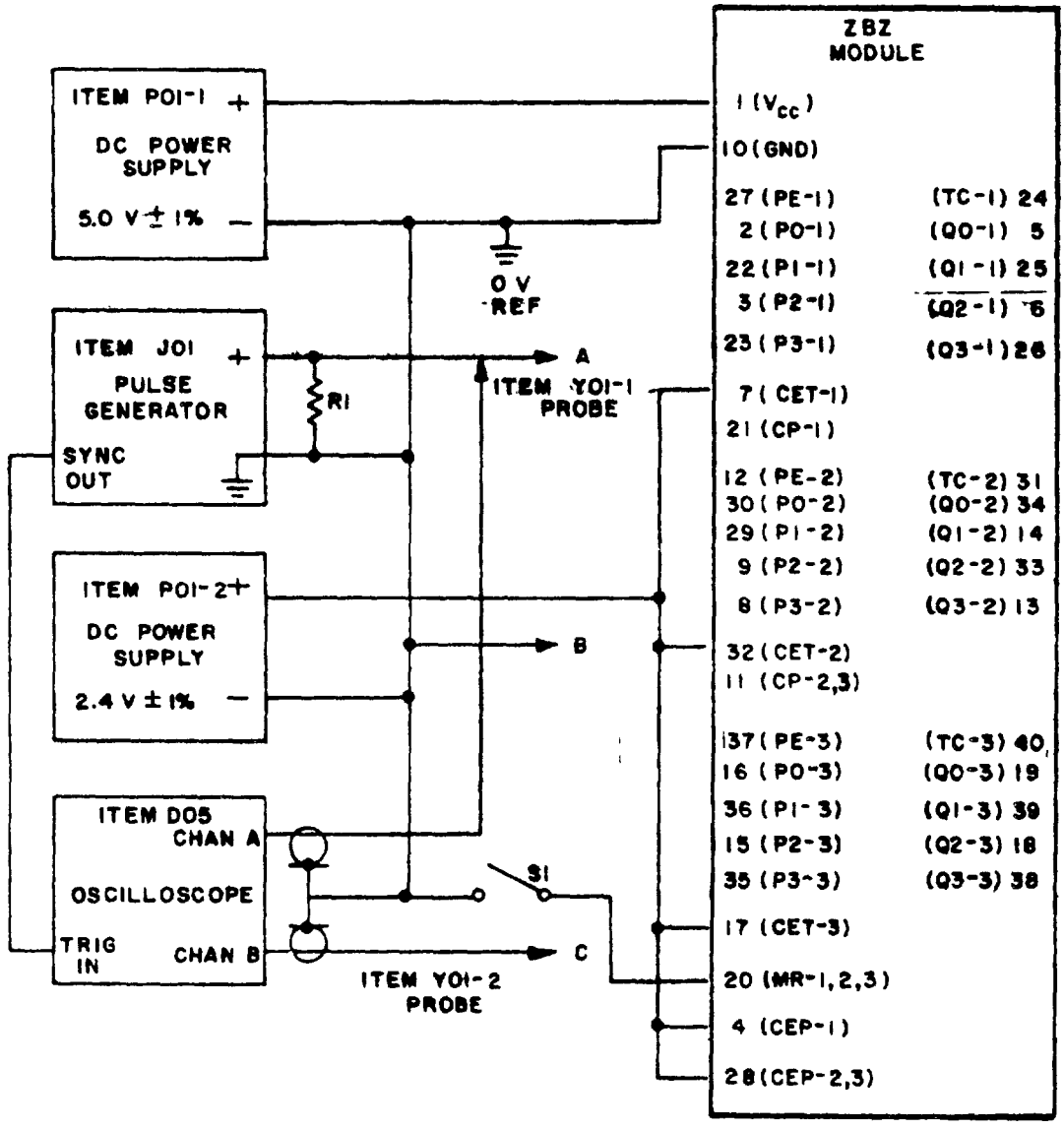
Test condition	Parameter	Connect				Initial limits 25° + 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
Y1	TC-1 Transition time (t_{TLH}), C_L = 40 pF	21	2,3,22	24	26 to 27	5	45	ns	16,18
Y2	TC-2	11	9,29,30	31	12 to 13				
Y3	TC-3	11	15,16,36	40	37 to 38				
Y4	QO-1	21	2,3,22	5	26 to 27				
Y5	Q3-1		23	26	5 to 27				16,17
Y6	Q1-1		3,22,23	25	6 to 27				16,17
Y7	Q2-1		3,23	6	25 to 27				16,17
Y8	QO-2	11	9,29,30	34	12 to 13				16,18
Y9	Q3-2		8	13	12 to 34				16,17
Y10	Q1-2		8,9,29	14	12 to 33				16,17
Y11	Q2-2		8,9	33	12 to 14				16,17
Y12	QO-3		15,16,36	19	37 to 38				16,18
Y13	Q3-3		35	38	19 to 37				16,17
Y14	Q1-3		15,35,36	39	18 to 37				16,17
Y15	Q2-3		15,35	18	37 to 39				16,17

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TABLE XVIII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° + 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
Z1	TC-1 Transition time (t_{TLH}), C_L = 100 pF	21	2,3,22	24	26 to 27	10	55	ns	16,18
Z2	TC-2	11	9,29,30	31	12 to 13				
Z3	TC-3	11	15,16,36	40	37 to 38				
Z4	Q0-1	21	2,3,22	5	26 to 27				
Z5	Q3-1		23	26	5 to 27				16,17
Z6	Q1-1		3,22,23	25	6 to 27				16,17
Z7	Q2-1		3,23	6	25 to 27				16,17
Z8	Q0-2	11	9,29,30	34	12 to 13				16,18
Z9	Q3-2		8	13	12 to 34				16,17
Z10	Q1-2		8,9,29	14	12 to 33				16,17
Z11	Q2-2		8,9	33	12 to 14				16,17
Z12	Q0-3		15,16,36	19	37 to 38				16,18
Z13	Q3-3		35	38	19 to 37				16,17
Z14	Q1-3		15,35,36	39	18 to 37				16,17
Z15	Q2-3		15,35	18	37 to 39				16,17

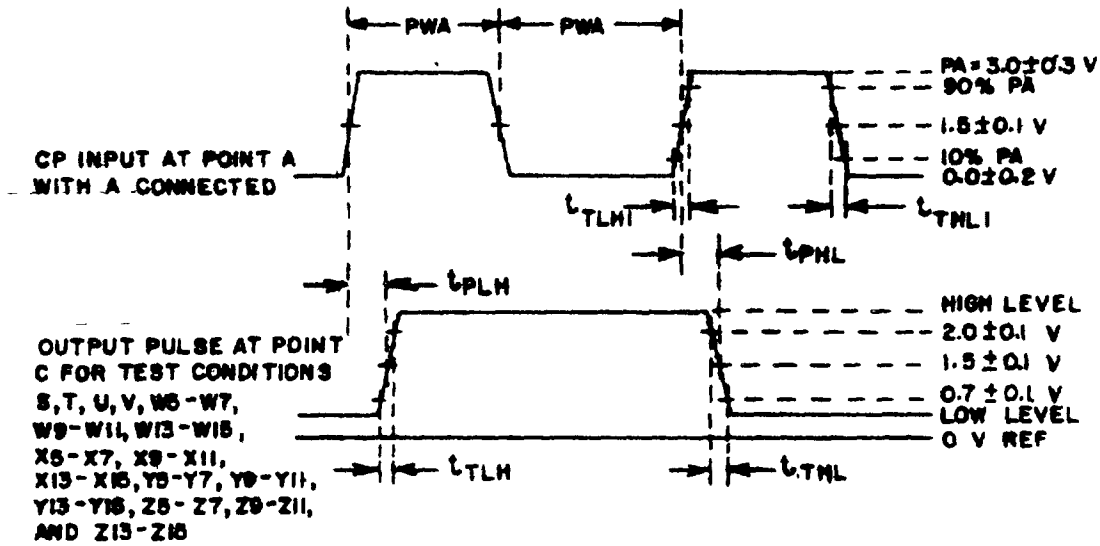
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Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST.
3. Load all outputs with a logic load circuit as specified on figure 10.
4. After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns. Then open switch S1 and take measurement.

FIGURE 16 Propagation delay time (t_{PLH} and t_{PHL}) and transition time (t_{TLH} and t_{THL}) test circuit for tables XV, XVI, XVII, and XVIII

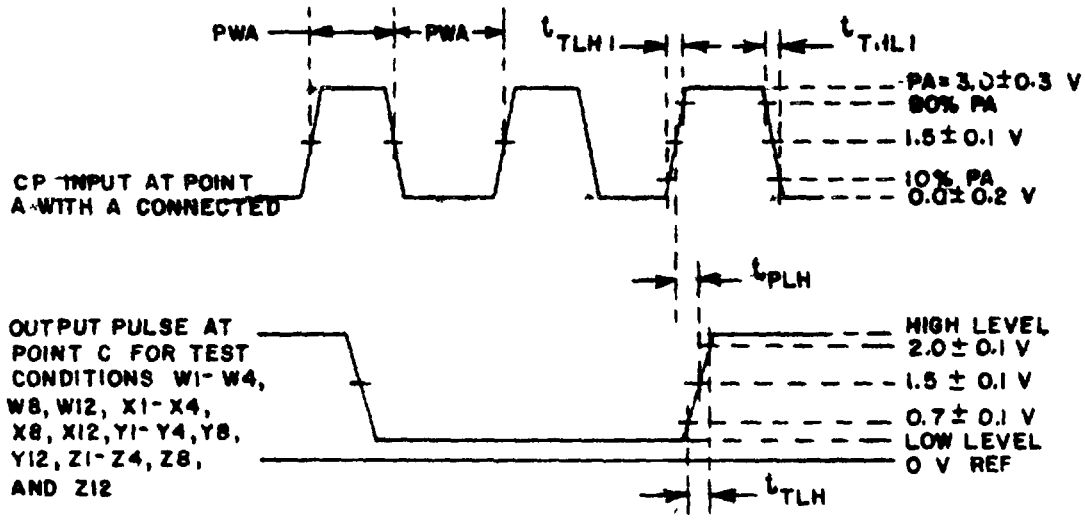
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:41Z
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NOTES:
1. Input pulse parameters:
   a. PWA = 500 ±50 ns.
   b. $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Before checking output pulse characteristics, verify that the low level is 0.3 V maximum and the high level is 2.4 V minimum.
3. All voltage levels are referenced to ground.

FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) and transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XV, XVI, XVII, and XVIII.

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NOTES:
1. Input pulse parameters:
   a. PWA = 500 ±50 ns.
   b. $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
2. Before checking output pulse characteristics, verify that the low level is 0.3 V maximum and the high level is 2.4 V minimum.
3. All voltage levels are referenced to ground.

FIGURE 18 Propagation delay time (t_{PLH}) and transition time (t_{TLH}) test circuit waveforms for tables XVII and XVIII.

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TABLE XIX. Master reset delay time (t_{PHL}) test.

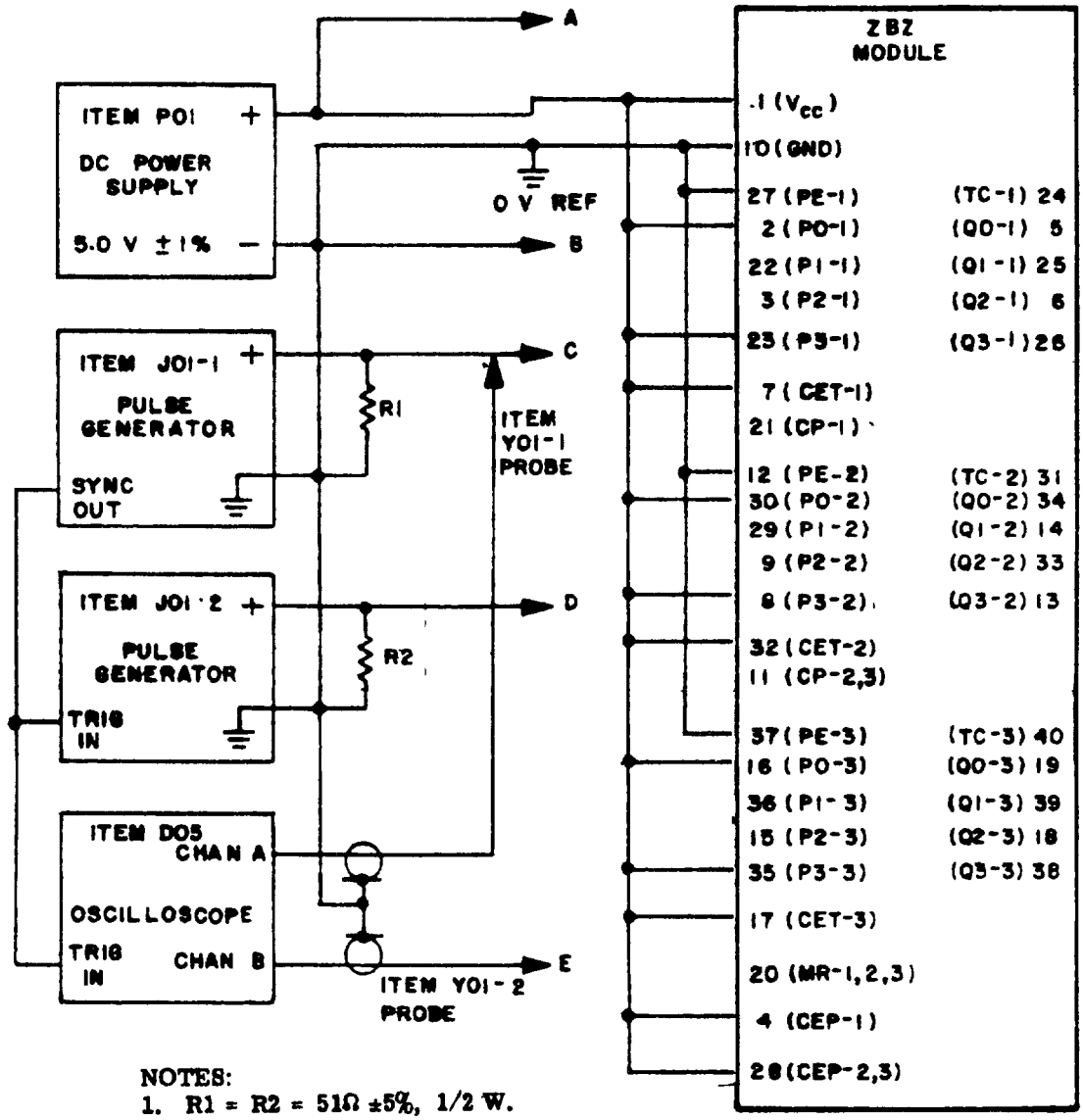
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
AA1	MR-1,2,3 to Q3-1 Master reset delay time (t_{PHL}), C_L = 40 pF	3,22	Open	20	21	26	25	99	ns	19,20
AA2	MR-1,2,3 to Q2-1					6				
AA3	MR-1,2,3 to Q1-1					25				
AA4	MR-1,2,3 to Q0-1					5				
AA5	MR-1,2,3 to TC-1	Open	3,22			24				
AA6	MR-1,2,3 to Q3-2	9,29	Open		11	13				
AA7	MR-1,2,3 to Q2-2					33				
AA8	MR-1,2,3 to Q1-2					14				
AA9	MR-1,2,3 to Q0-2					34				
AA10	MR-1,2,3 to TC-2	Open	9,29			31				
AA11	MR-1,2,3 to Q3-3	15,36	Open			38				
AA12	MR-1,2,3 to Q2-3					18				
AA13	MR-1,2,3 to Q1-3					39				
AA14	MR-1,2,3 to Q0-3					19				
AA15	MR-1,2,3 to TC-3	Open	15,36			40				

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TABLE XIX Master reset delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	MR-1,2,3 to Q3-1 Master reset delay time (t_{PHL}), C_L = 100 pF	3,22	Open	20	21	26	25	105	ns	19,20
BB2	MR-1,2,3 to Q2-1					6				
BB3	MR-1,2,3 to Q1-1					25				
BB4	MR-1,2,3 to Q0-1					5				
BB5	MR-1,2,3 to TC-1	Open	3,22			24				
BB6	MR-1,2,3 to Q3-2	9,29	Open		11	13				
BB7	MR-1,2,3 to Q2-2					33				
BB8	MR-1,2,3 to Q1-2					14				
BB9	MR-1,2,3 to Q0-2					34				
BB10	MR-1,2,3 to TC-2	Open	9,29			31				
BB11	MR-1,2,3 to Q3-3	15,36	Open			38				
BB12	MR-1,2,3 to Q2-3					18				
BB13	MR-1,2,3 to Q1-3					39				
BB14	MR-1,2,3 to Q0-3					19				
BB15	MR-1,2,3 to TC-3	Open	15,36			40				

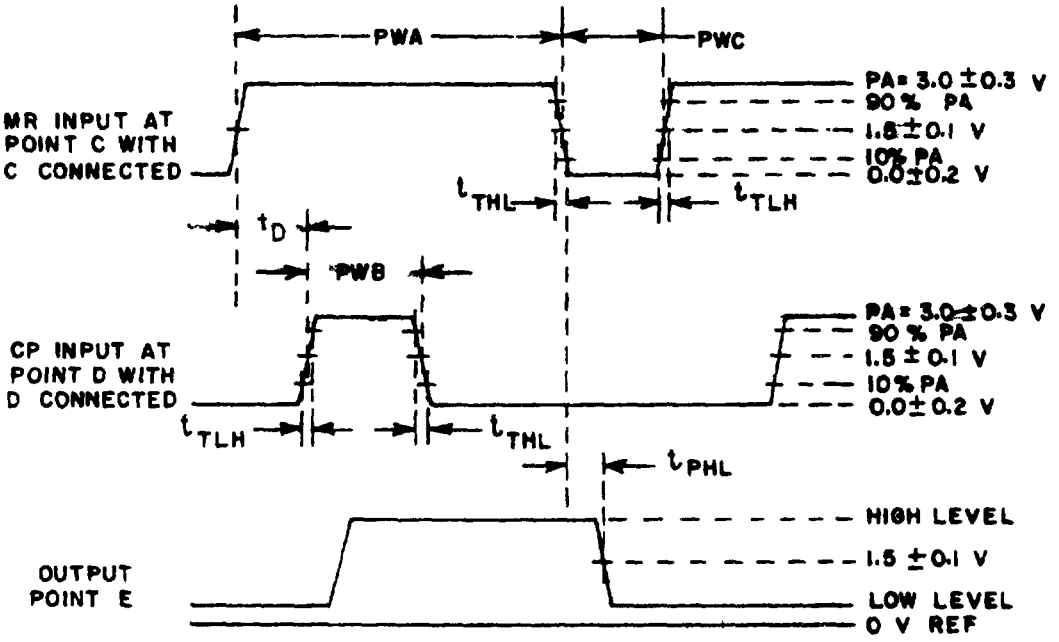
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a logic load circuit as specified on figure 10.

FIGURE 19. Master reset delay time (t_{PHL}) test circuit for table XIX.

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NOTES

1. Input pulse parameters:

   a. PWA = 300 ±30 ns.
   b. PWB = 100 ±10 ns.
   c. PWC = 65 ±2 ns (minimum MR pulse width).
   d. t_D = 50 ±2 ns (minimum MR recovery time).
   e. $t_{TLH} = t_{THL}$ = 10 ±2 ns.

2. All voltage levels are referenced to ground.

FIGURE 20 Master reset delay time (t_{PHL}) test circuit waveforms for table XIX.

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TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) test.

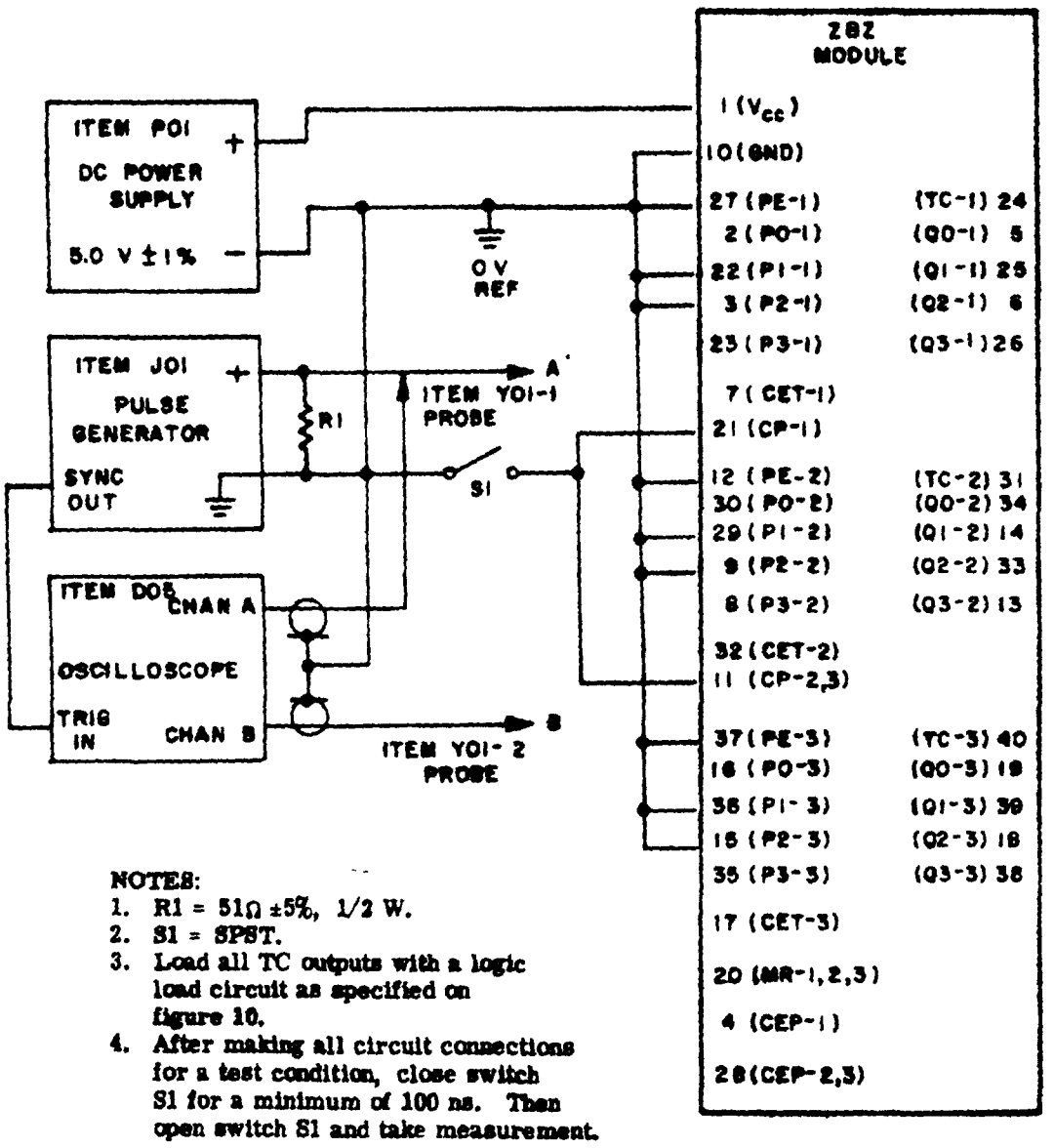
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	CET-1 to TC-1 Propagation delay time (t_{PHL}), C_L = 40 pF	7	24	10	64	ns	21,22
CC2	CET-2 to TC-2 Propagation delay time (t_{PHL}), C_L = 40 pF	32	31		64		
CC3	CET-3 to TC-3 Propagation delay time (t_{PHL}), C_L = 40 pF	17	40		64		
CC4	CET-1 to TC-1 Propagation delay time (t_{PLH}), C_L = 40 pF	7	24		59		
CC5	CET-2 to TC-2 Propagation delay time (t_{PLH}), C_L = 40 pF	32	31		59		
CC6	CET-3 to TC-3 Propagation delay time (t_{PLH}), C_L = 40 pF	17	40		59		
DD1	CET-1 to TC-1 Propagation delay time (t_{PHL}), C_L = 100 pF	7	24	15	70		
DD2	CET-2 to TC-2 Propagation delay time (t_{PHL}), C_L = 100 pF	32	31	15	70		

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TABLE XX. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
DD3	CET-3 to TC-3 Propagation delay time (t_{PHL}), C_L = 100 pF	17	40	15	70	ns	21,22
DD4	CET-1 to TC-1 Propagation delay time (t_{PLH}), C_L = 100 pF	7	24		65		
DD5	CET-2 to TC-2 Propagation delay time (t_{PLH}), C_L = 100 pF	32	31		65		
DD6	CET-3 to TC-3 Propagation delay time (t_{PLH}), C_L = 100 pF	17	40		65		

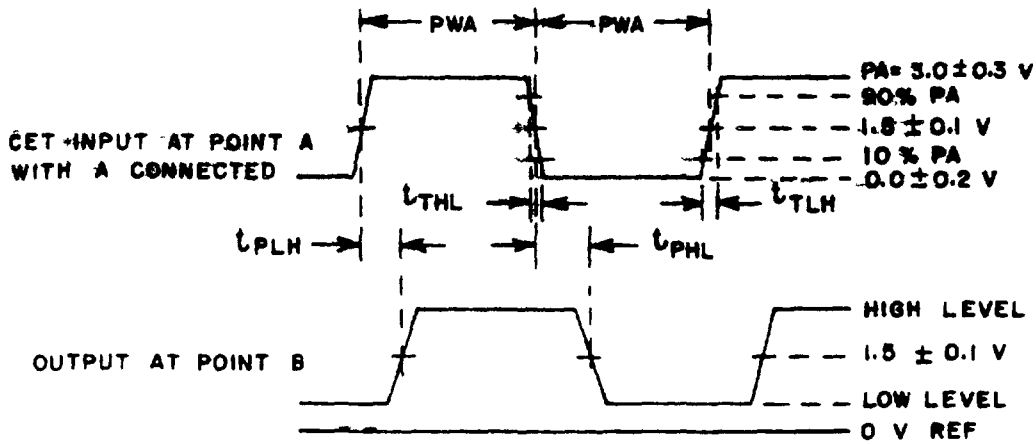
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST.
3. Load all TC outputs with a logic load circuit as specified on figure 10.
4. After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns. Then open switch S1 and take measurement.

FIGURE 21. Propegation delay time (t_{PLH} and t_{PHL}) test circuit for table XX

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NOTES:
1. Input pulse parameters

   a. PWA = 500 ±50 ns.
   b. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.

2. All voltage levels are referenced to ground.

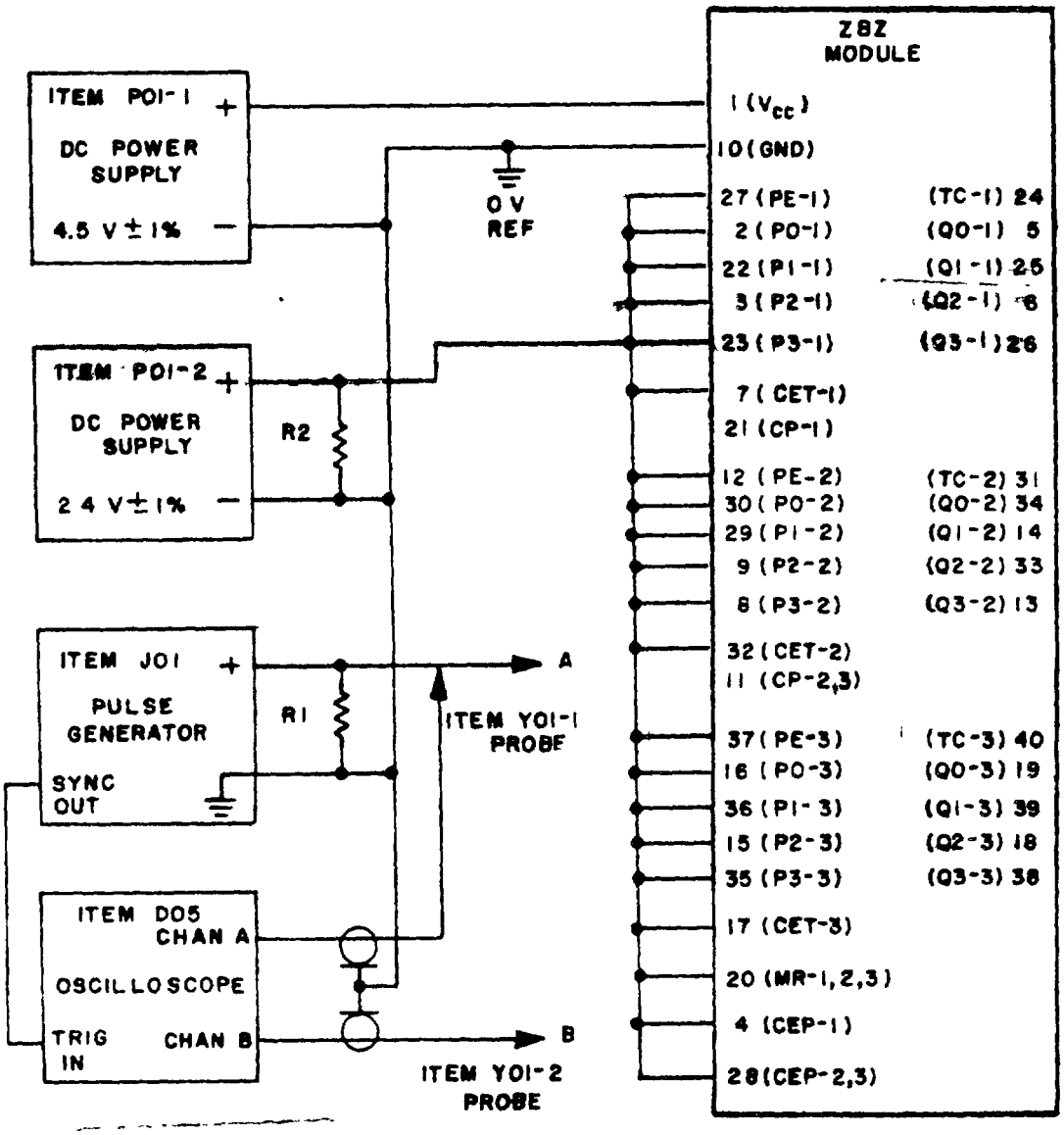
FIGURE 22 Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XX

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TABLE XXI. Maximum counting frequency check test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
EE1	PW1 Maximum counting frequency check	21	26	225	275	ns	23,24
EE2	PW1	11	13	225	275		
EE3	PW1	11	38	225	275		
EE4	PW2	21	26	975	1,025		
EE5	PW2	11	13	975	1,025		
EE6	PW2	11	38	975	1,025		

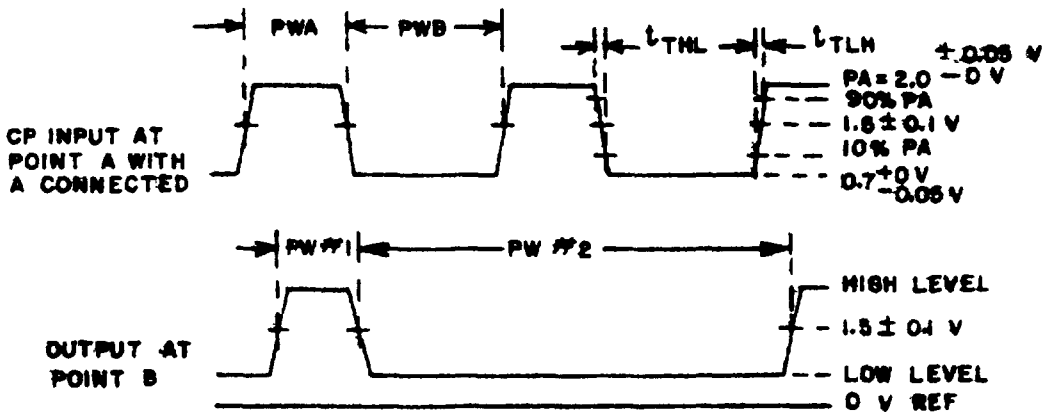
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a logic load circuit as specified on figure 10.

FIGURE 23. Maximum counting frequency check test circuit for table XXI.

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NOTES:
1. Input pulse parameters:
   a. PWA = 50 $^{+2}_{-0}$ ns (minimum CP pulse width).
   b. PWB = 75 $^{+5}_{-0}$ ns (minimum CP pulse space).
   c. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. All voltage levels are referenced to ground.
3. This test verifies maximum counting frequency of 1.2 MHz under threshold clock conditions.

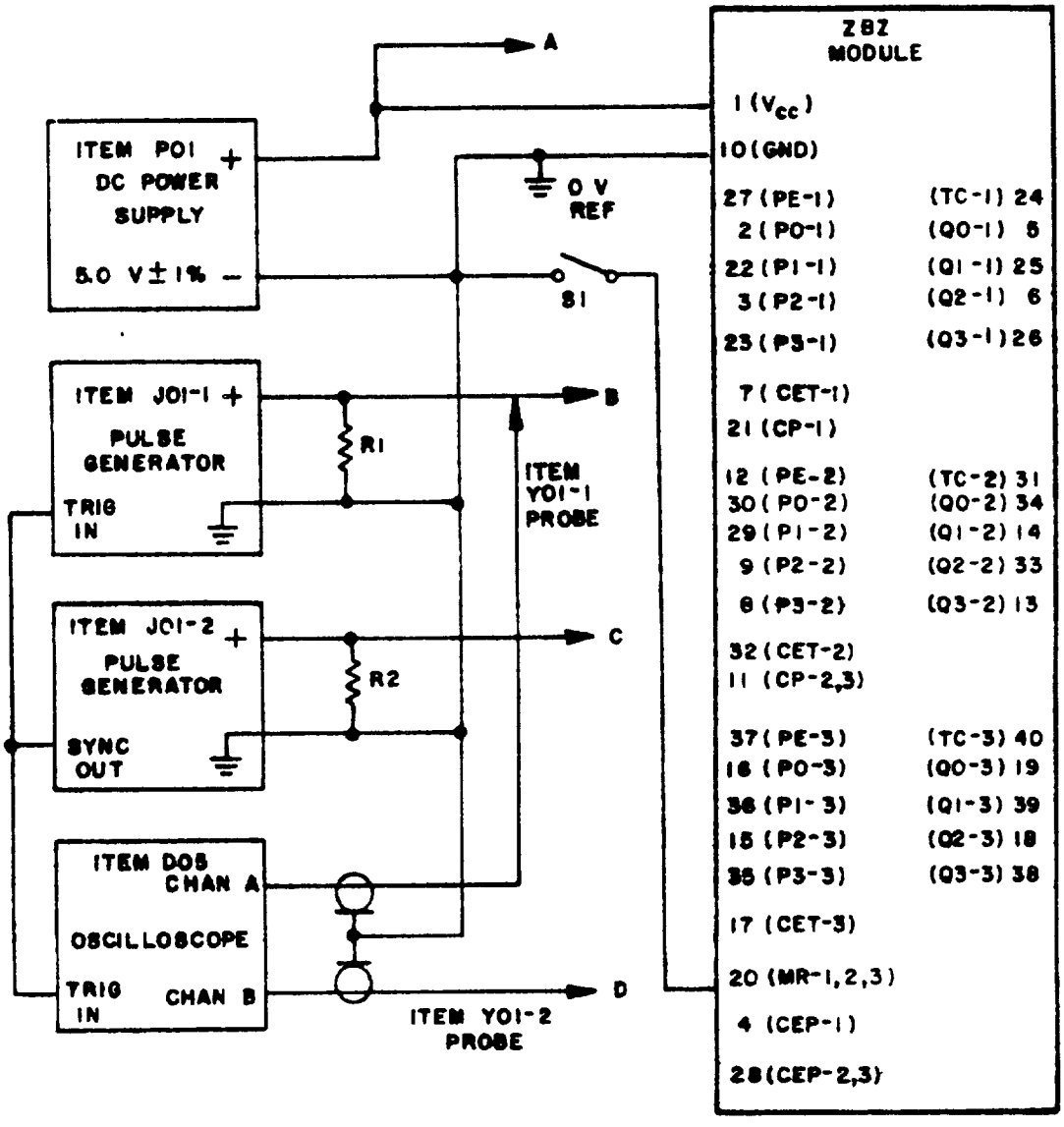
FIGURE 24. Maximum counting frequency check test circuit waveforms for table XXI.

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TABLE XXII. Setup and hold time check test

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
FF1	CET-1 Setup and hold time check (output period)	4	7	21	26	1.9	2.1	µs	25,26
FF2	CET-2	28	32	11	13				
FF3	CET-3	28	17	11	38				
FF4	CEP-1	7	4	21	26				
FF5	CEP-2,3	32	28	11	13				
FF6	CEP-2,3	17	28	11	38				

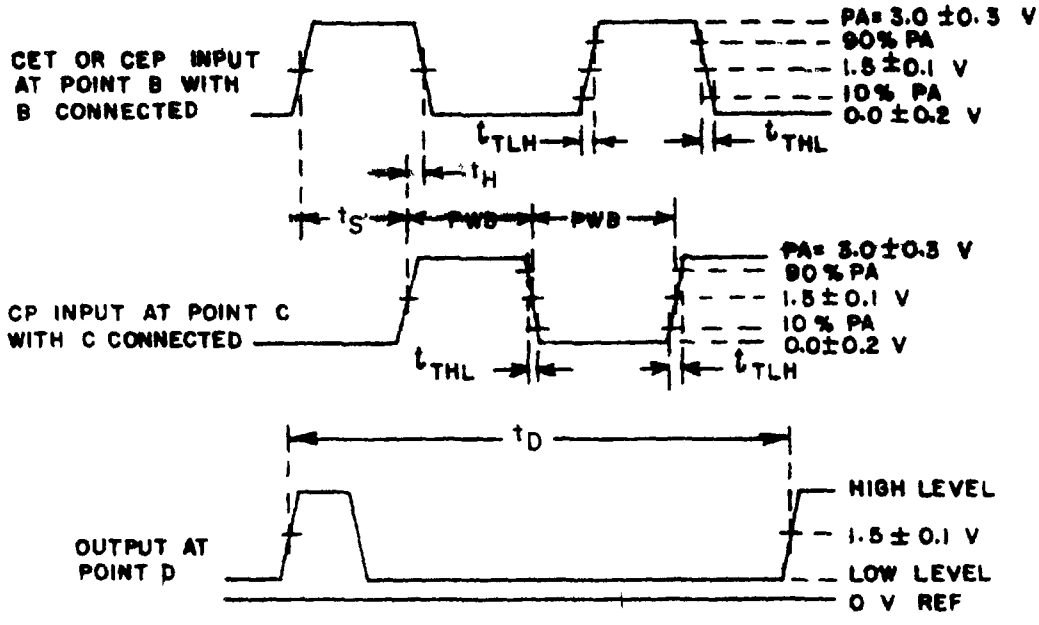
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. S1 = SPST.
3. Load all Q3 outputs with a logic load circuit as specified on figure 10.
4. After making all connections for a test condition, close switch S1 for a minimum of 100 ns. Then open switch S1 and take measurement.

FIGURE 25. Setup and hold time check test circuit for table XXII.

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NOTES:

1 Input pulse parameters·

a. $t_S = 80 \, ^{+4}_{-0}$ ns

b $t_H = 0 \, ^{+2}_{-0}$ ns

c. PWB = 100 ±5 ns.

d. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.

2. Output waveform D time scale is not the same as point B and C.

3. All voltage levels are referenced to ground.

FIGURE 26 Setup and hold time check test circuit waveforms for table XXII.

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TABLE XXIII. Setup and hold time check test.

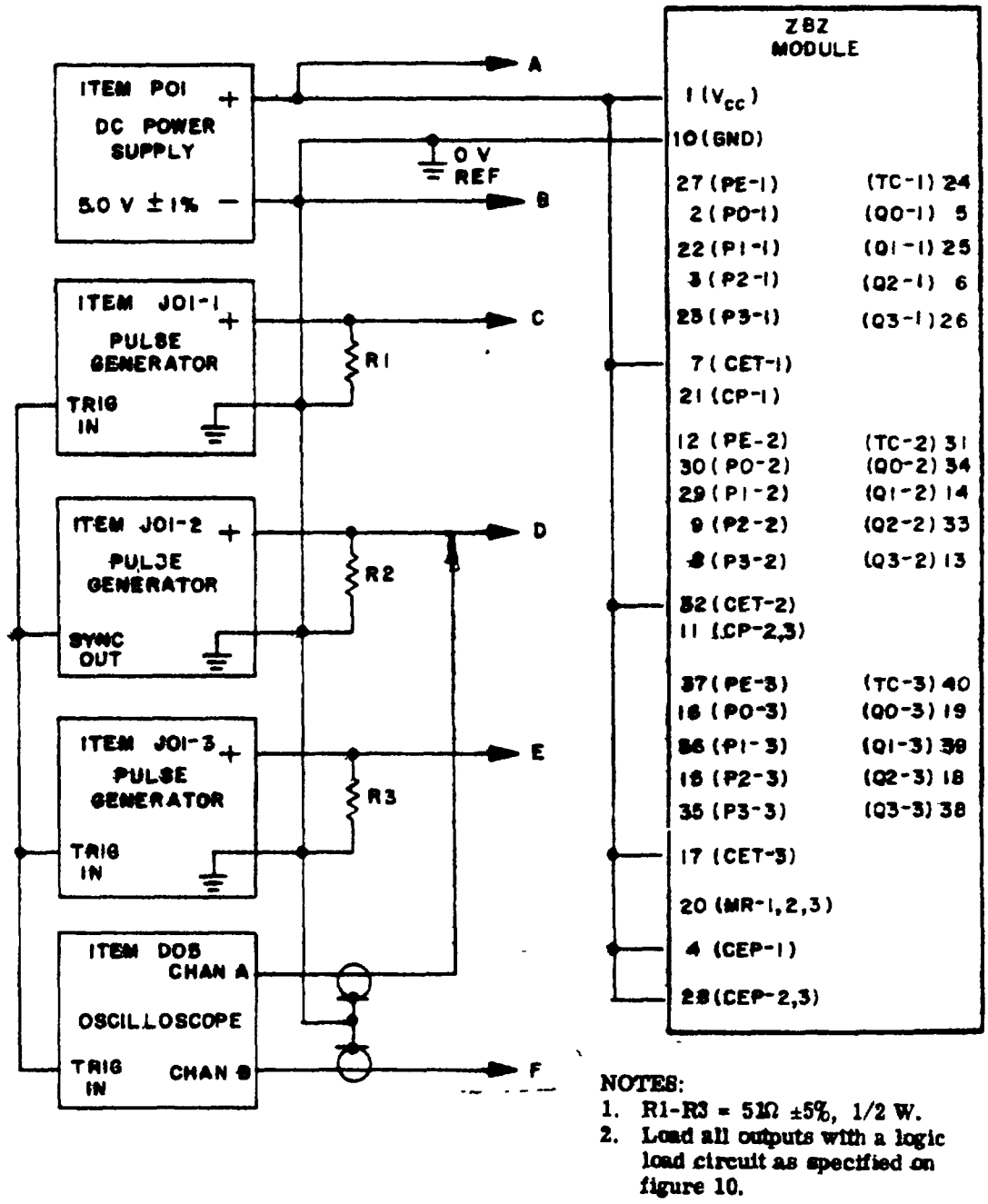
Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
GG1	PO-1 Setup and hold time check, high level	Open	27	2	21	20	5	10	54	ns	27,29
GG2	P1-1			22			25				
GG3	P2-1			3			6				
GG4	P3-1			23			26				
GG5	PO-2		12	30	11		34				
GG6	P1-2			29			14				
GG7	P2-2			9			33				
GG8	P3-2			8			13				
GG9	PO-3		37	16			19				
GG10	P1-3			36			39				
GG11	P2-3			15			18				
GG12	P3-3			35			38				
GG13	PE-1 Setup and hold time check, low level	23	Open	27	21		26		59		27,30
GG14	PE-2 Setup and hold time check, low level	8	Open	12	11		13		59		27,30
GG15	PE-3 Setup and hold time check, low level	35	Open	37	11		38		59		27,30

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TABLE XXIII. Setup and hold time check test - Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
GG16	P0-1 Setup and hold time check, low level	Open	Open	27	2	21	5	10	59	ns	28,31
GG17	P1-1				22		25				
GG18	P2-1				3		6				
GG19	P3-1				23		26				
GG20	P0-2			12	30	11	34				
GG21	P1-2				29		14				
GG22	P2-2				9		33				
GG23	P3-2				8		13				
GG24	P0-3			37	16		19				
GG25	P1-3				36		39				
GG26	P2-3				15		18				
GG27	P3-3				35		38				

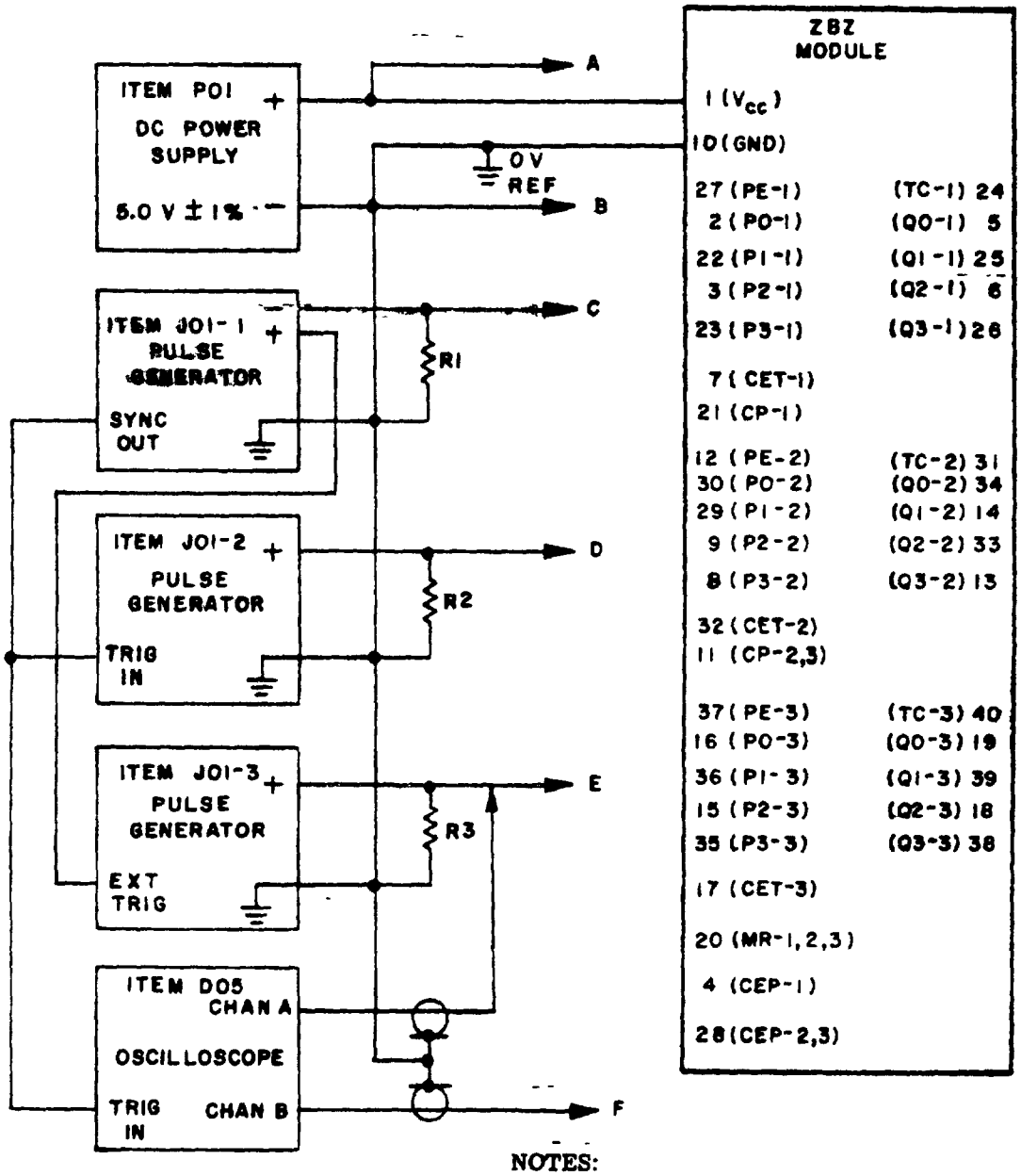
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a logic load circuit as specified on figure 10.

FIGURE 27. Setup and hold time check test circuit for table XXIII, test conditions GG1-GG15.

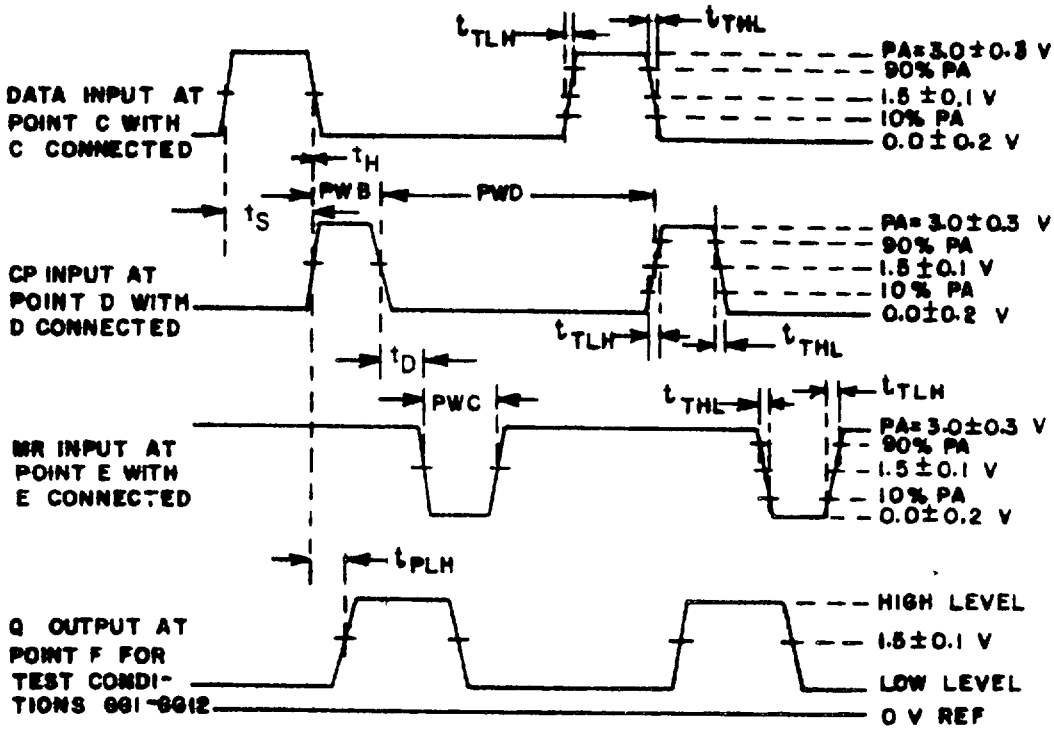
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a logic load circuit as specified on figure 10.
3. Place pulse generators J01-1 and J01-3 in the double pulse mode.

FIGURE 28. Setup and hold time check test circuit for table XXIII, test conditions GG16-GG27.

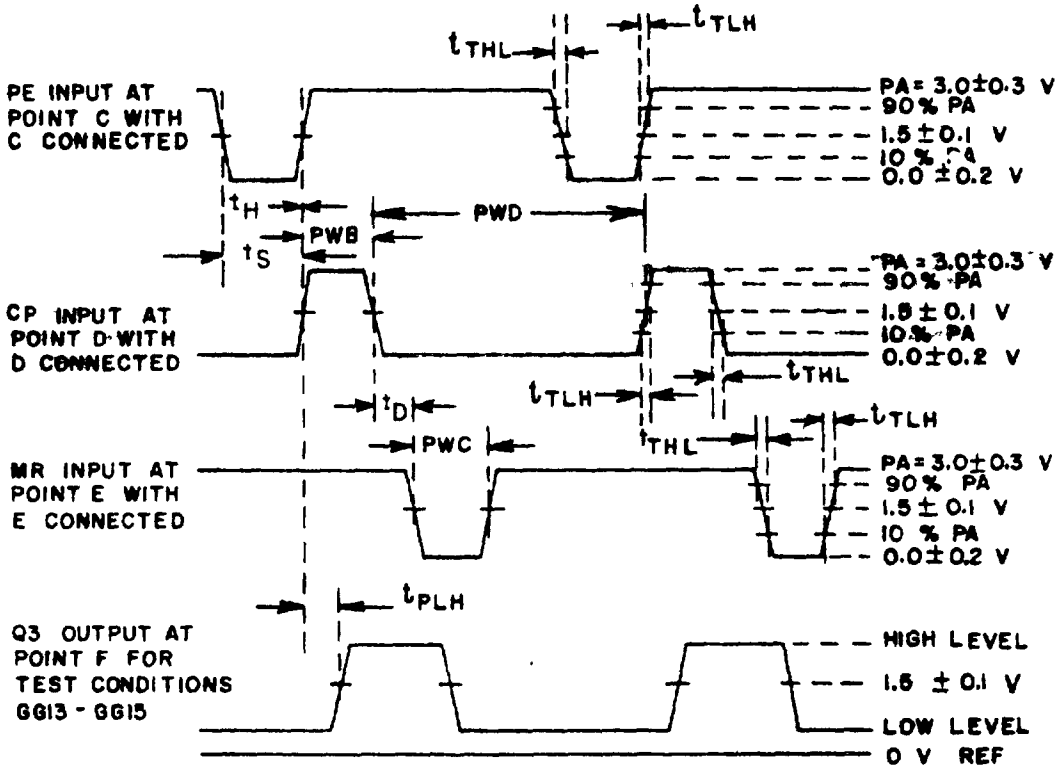
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:41Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   a. PWB = 300 ±30 ns.
   b. PWC = 100 ±10 ns.
   c. PWD = 600 ±30 ns.
   d. $t_S = 110 \, ^{+5}_{-0}$ ns.
   e. $t_H = 0 \, ^{+2}_{-0}$ ns.
   f. t_D = 100 ±10 ns.
   g. $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.
3. t_{PLH} shall be as specified in table XXIII.

FIGURE 29. Setup and hold time check test circuit waveforms for table XXIII.

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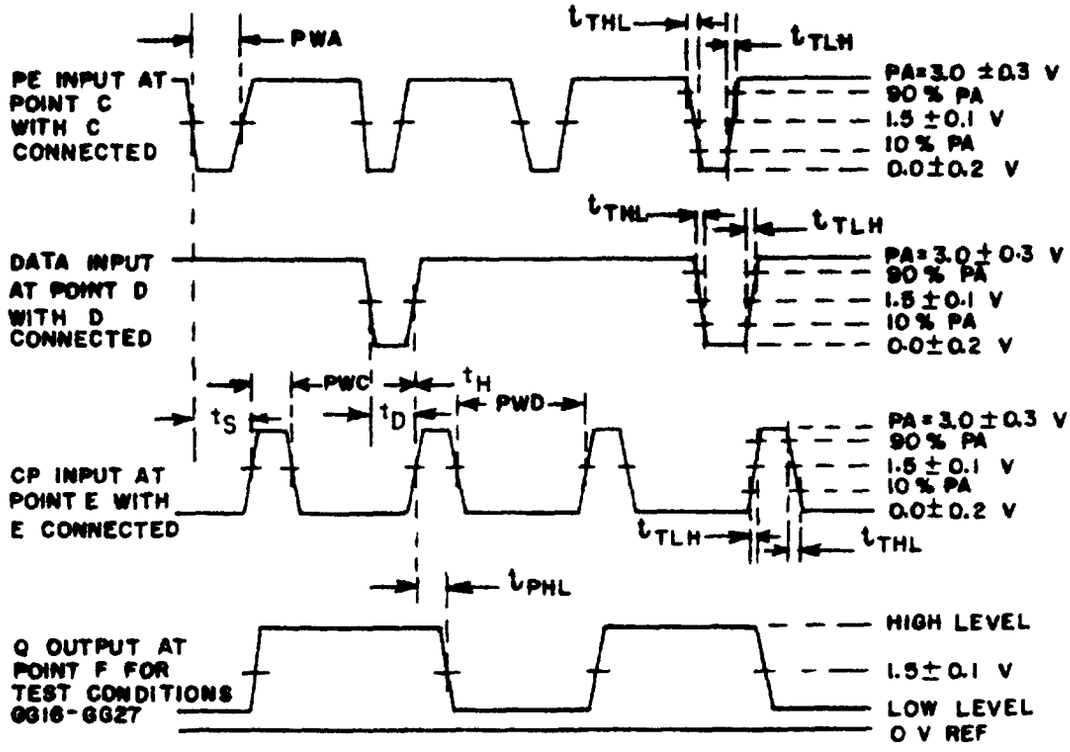
NOTES:
1. Input pulse parameters:

   a. PWB = 300 ±30 ns.
   b. PWC = 100 ±10 ns.
   c. PWD = 600 ±30 ns.
   d. $t_S = 80 \begin{smallmatrix}+4\\-0\end{smallmatrix}$ ns.
   e. $t_H = 0 \begin{smallmatrix}+2\\-0\end{smallmatrix}$ ns.
   f. t_D = 100 ±10 ns.
   g. $t_{TLH} = t_{THL}$ = 10 ±2 ns.

2. All voltage levels are referenced to ground.
3. t_{PLH} shall be as specified in table XXIII.

FIGURE 30. Setup and hold time check test circuit waveforms for table XXIII.

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NOTES:

1. Input pulse parameters:

   a. PWA = 150 ±10 ns.
   b. PWC = 300 ±30 ns.
   c. PWD = 300 ±30 ns.
   d. $t_S = 110 \, ^{+5}_{-0}$ ns.
   e. $t_H = 0 \, ^{+2}_{-0}$ ns.
   f. t_D = 125 ±10 ns.
   g. $t_{TLH} = t_{THL}$ = 10 ±2 ns.

2. All voltage levels are referenced to ground.
3. t_{PHL} shall be as specified in table XXIII.

FIGURE 31. Setup and hold time check test circuit waveforms for table XXIII.

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TABLE XXIV Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
HH1	Q0-1 Output crosstalk, high level	2	3,8,9, 15,16, 22,23, 29,30, 35,36	Open	5	0	400	mV	32,33
HH2	Q1-1	22	2,3,8, 9,15, 16,23, 29,30, 35,36		25				
HH3	Q2-1		2,8,9, 15,16, 22,23, 29,30, 35,36		6				
HH4	Q3-1	23	2,3,8, 9,15, 16,22, 29,30, 35,36		26				
HH5	TC-1	2,23	8,9,15, 16,29, 30,35, 36	3,22	24				
HH6	Q0-2	30	2,3,8, 9,15, 16,22, 23,29, 35,36	Open	34				

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TABLE XXIV Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
HH7	Q1-2 Output crosstalk, high level	29	2,3,8, 9,15, 16,22, 23,30, 35,36	Open	14	0	400	mV	32,33
HH8	Q2-2	9	2,3,8, 15,16, 22,23, 29,30, 35,36	Open	33				
HH9	Q3-2	8	2,3,9, 15,16, 22,23, 29,30, 35,36	Open	13				
HH10	TC-2	8,30	2,3,15, 16,22, 23,35, 36	9,29	31				
HH11	Q0-3	16	2,3,8, 9,15, 22,23, 29,30, 35,36	Open	19				
HH12	Q1-3	36	2,3,8, 9,15, 16,22, 23,29, 30,35	Open	39				

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TABLE XXIV Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
HH13	Q2-3 Output crosstalk, high level	15	2,3,8, 9,16, 22,23, 29,30, 35,36	Open	18	0	400	mV	32,33
HH14	Q3-3 Output crosstalk, high level	35	2,3,8, 9,15, 16,22, 23,29, 30,36	Open	38		400		
HH15	TC-3 Output crosstalk, high level	16, 3	2,3,8, 9,22, 23,29, 30	15,36	40		400		
JJ1	QO-1 Output crosstalk, low level	Open	3,8,9, 15,16, 22,23, 29,30, 35,36	2	5		500		
JJ2	Q1-1 Output crosstalk, low level	Open	2,3,8, 9,15, 16,23, 29,30, 35,36	22	25		500		
JJ3	Q2-1 Output crosstalk, low level	Open	2,8,9, 15,16, 22,23, 29,30, 35,36	3	6				

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TABLE XXIV. Output crosstalk (low-level and high-level) test - Continued.

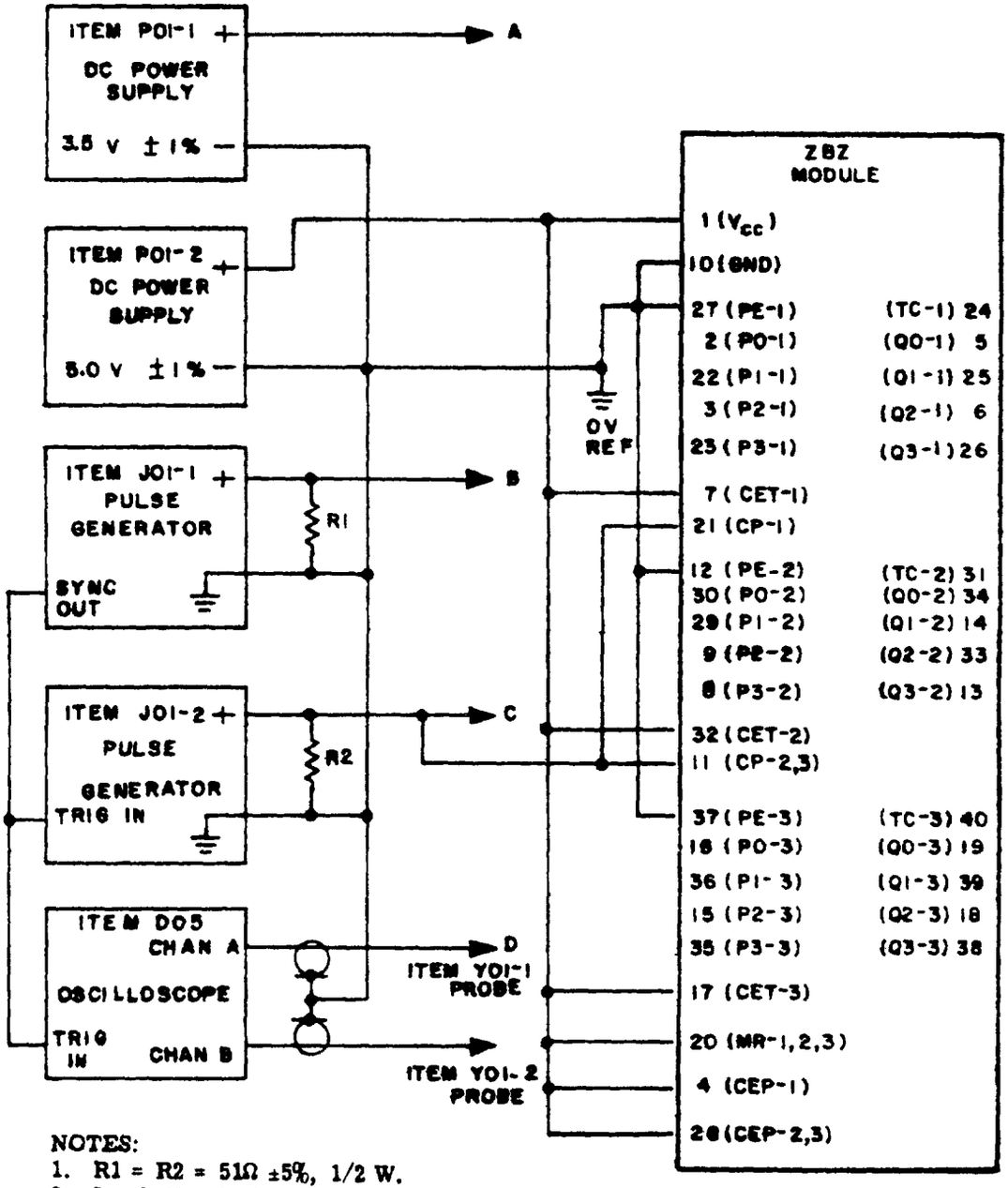
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
JJ4	Q3-1 Output crosstalk, low level	Open	2,3,8, 9,15, 16,22, 29,30, 35,36	23	26	0	500	mV	32,33
JJ5	TC-1		2,3,8, 9,15, 16,22, 29,30, 35,36	23	24				
JJ6	Q0-2		2,3,8, 9,15, 16,22, 23,29, 35,36	30	34				
JJ7	Q1-2		2,3,8, 9,15, 16,22, 23,30, 35,36	29	14				
JJ8	Q2-2		2,3,8, 15,16, 22,23, 29,30, 35,36	9	33				
JJ9	Q3-2		2,3,9, 15,16, 22,23, 29,30, 35,36	8	13				

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TABLE XXIV Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
JJ10	TC-2 Output crosstalk, low level	Open	2,3,9, 15,16, 22,23, 29,30, 35,36	8	31	0	500	mV	32,33
JJ11	Q0-3		2,3,8, 9,15, 22,23, 29,30, 35,36	16	19				
JJ12	Q1-3		2,3,8, 9,15, 16,22, 23,29, 30,35	36	39				
JJ13	Q2-3		2,3,8, 9,16, 22,23, 29,30, 35,36	15	18				
JJ14	Q3-3		2,3,8, 9,15, 16,22, 23,29, 30,36	35	38				
JJ15	TC-3		2,3,8, 9,15, 16,22, 23,29, 30,36	35	40				

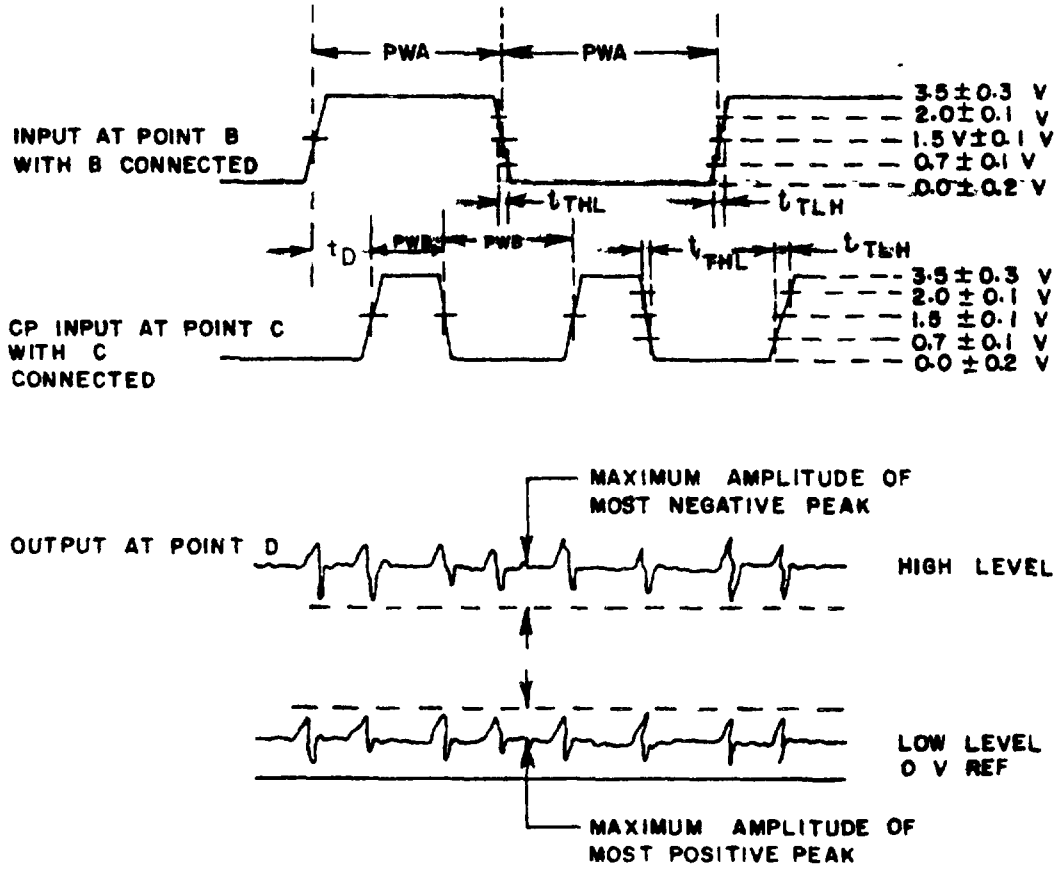
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a logic load circuit as specified on figure 10.

FIGURE 32. Output crosstalk (low-level and high-level) test circuit for table XXIV.

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NOTE: Input pulse parameters:

a. PWA = 500 ±50 ns.
b. PWB = 250 ±25 ns.
c. t_D = 250 ±50 ns.
d. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 33. Output crosstalk (low-level and high-level) test circuit waveforms for table XXIV.

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TABLE XXV. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
KK	Power supply filter capacitance	1.6	3.0	µF	14

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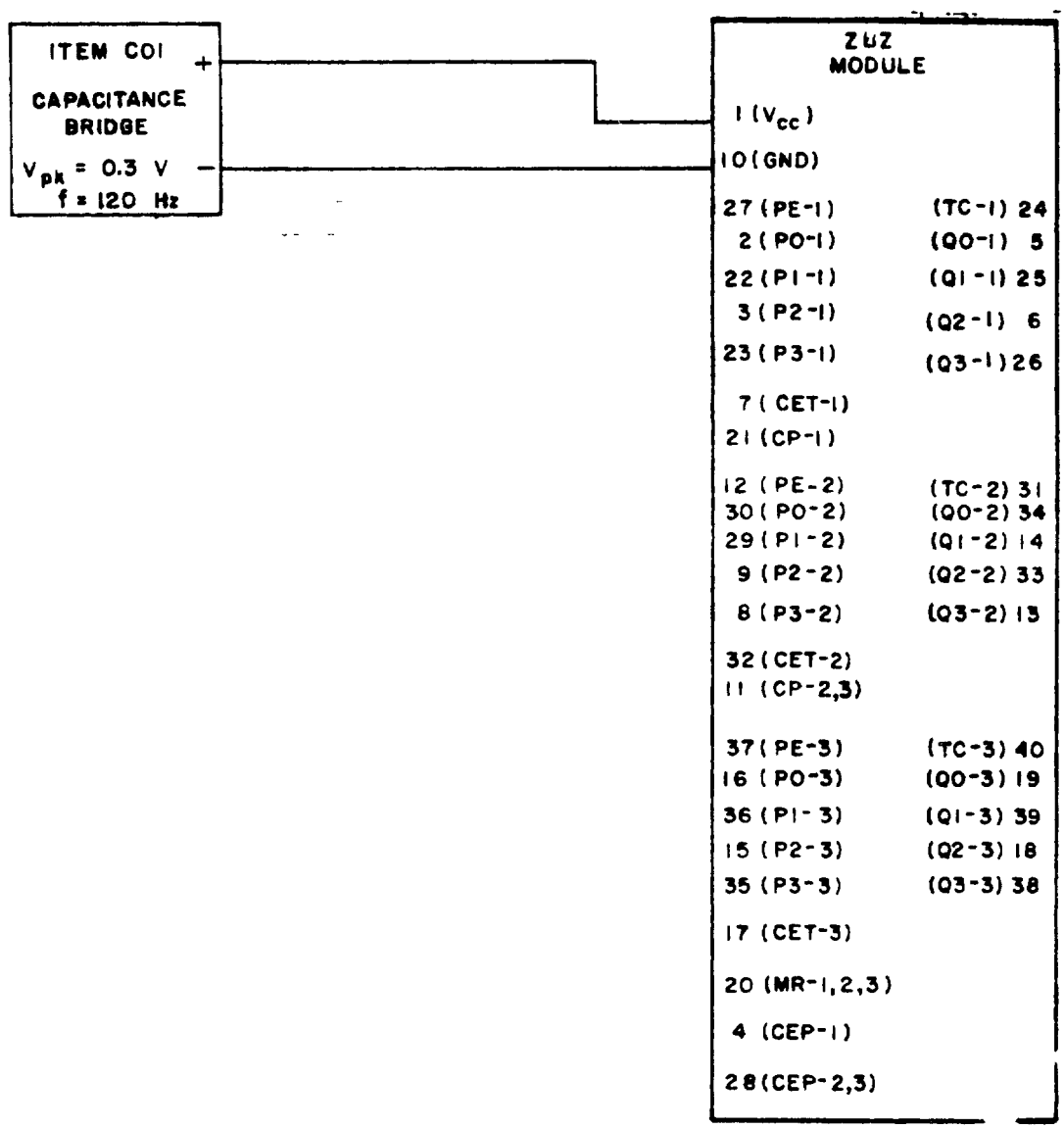


FIGURE 34. Power supply filter capacitance test circuit for table XXV.

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